Informatica® Cloud Data Governance and Catalog

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Data Access Management

Informatica Cloud Data Governance and Catalog Data Access Management
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Preface

Read *Data Access Management* to learn how to use data access policies to control access to data and apply data protections to it.

CHAPTER 1

Introduction to Data Access Management

Use the features on the Data Access Management page to build policy-based data access controls and protections into data operations, which enable efficient, effective, and responsible data use.

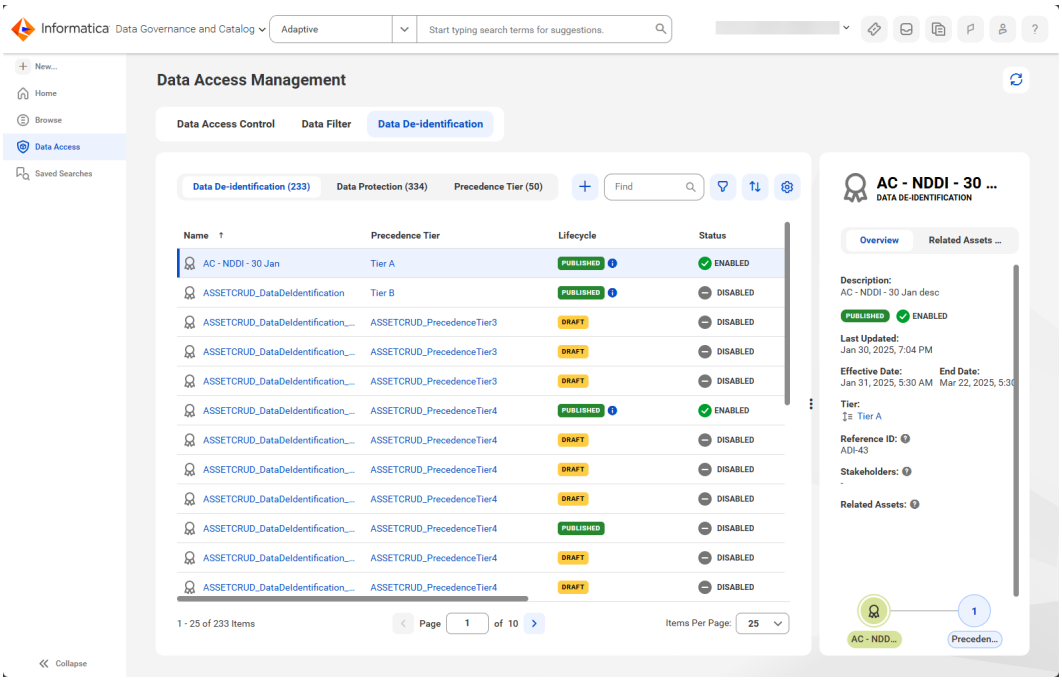
You can use the Data Access Management page to develop policies that apply in the following contexts:

- Through data operations in Data Integration and Data Ingestion and Replication
- Through self-service access in Data Marketplace
- Through direct enforcement in your source systems

In all cases, to see the data, you need to use appropriate tools relevant to that source system.

The policies that you build use metadata conditions to transform and deliver data in a way that satisfies the data protection and safety criteria of your organization.

The following image shows the Data Access Management page:



How data access policies and rules protect your data

Policies and rules protect your data by controlling access to it and applying data de-identification techniques that mask sensitive data while maintaining the data's usefulness. In Data Governance and Catalog, you can create reusable policies based on user, usage, or metadata context. These policies execute automatically on an unlimited number of data access requests.

Data Governance and Catalog offers data de-identification policies, data filter policies, and data access control policies.

- You use data de-identification policies to define data protections to apply to data element classifications to de-identify them for a given use case.
- You use data filter policies to filter access to rows within data assets.
- You use data access control policies to grant groups of users read, write, or delete access to tables or views in a source system. Data Governance and Catalog pushes these policies down into your cloud data platform, which enforces the policy directly.

You can reuse policies across different environments or for multiple data releases to ensure consistent treatment for the same data asset.

How to de-identify data while maintaining its usefulness

Using data de-identification policies as an example, let's start by looking at the result you want to achieve and see how Data Governance and Catalog helps you achieve it.

Your goal is to de-identify data while still maintaining that data's usefulness. This is where Data Governance and Catalog's data protections come in. A data protection defines a data de-identification technique for Data Governance and Catalog to apply to a given data element in a data set to de-identify it, while still preserving utility.

However, you don't want Data Governance and Catalog to apply the same data de-identification techniques to every data element in every data asset regardless of the user, the type of data, and the usage context. You want these data de-identification techniques to be conditional on the usage context. You use rules to ensure that Data Governance and Catalog applies the relevant data de-identification techniques to the right data elements according to the conditions of the data access request.

Rules are the building blocks of policies. Rules are conditional based on attributes such as user, characteristics of the data, and the usage context. Rules also take actions specific to data classes and data protections.

While rules apply to specific conditions, you might need multiple rules to meet the needs of a broader use case. You need to group this set of rules into a policy that defines the use case for which Data Governance and Catalog applies them.

Similarly, you group related data de-identification policies into precedence tiers.

Data access policy enforcement methods

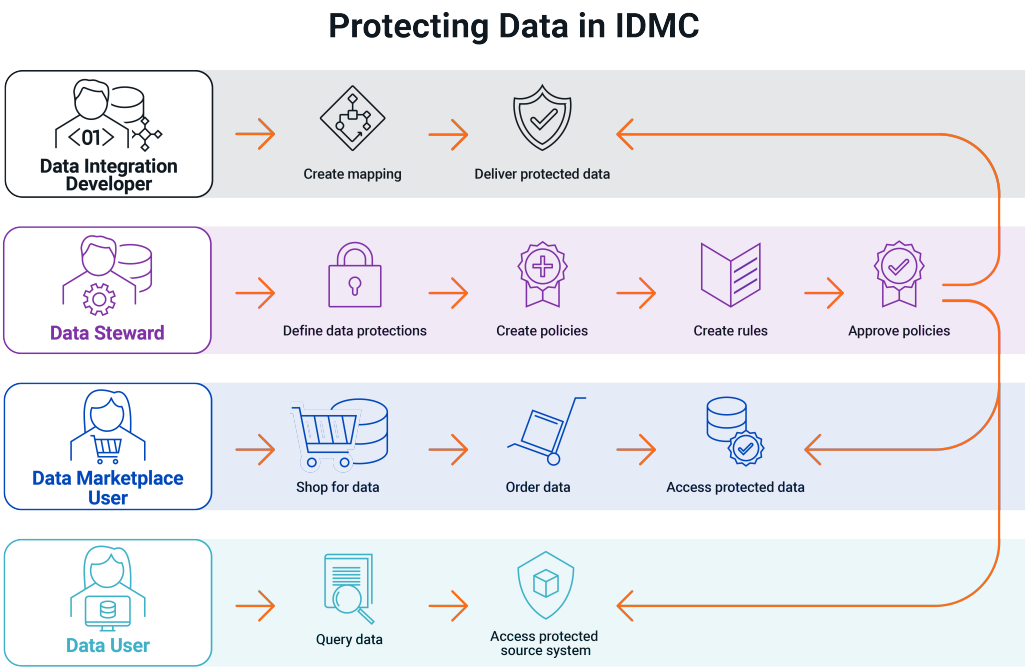
Data Access Management works with other Informatica services to protect data and provide access to it. As a Data Steward, it's important that you know where enforcement occurs for each type of data access policy that you create in Data Access Management.

The data access policies you create are enforced in the following contexts:

- Through data operations in Data Integration and Data Ingestion and Replication
- Through self-service access in Data Marketplace
- Through direct enforcement in your source systems

In all cases, to see the data, you need to use appropriate tools relevant to that source system.

The following image shows how policies that you create in Data Access Management protect data:



When you use Data Integration to build a data pipeline, you can connect an Access Policy transformation to the mapping. The Access Policy transformation applies data filter policies and data de-identification policies created in Data Access Management according to the properties of the Access Policy transformation.

In Data Marketplace, when you enable the **Managed Access** setting to place an order, Data Marketplace enforces data filter policies and data de-identification policies created in Data Access Management according to the usage specified in the Data Marketplace order.

For more information about placing orders, see the *Set Up Data Marketplace* guide in Data Marketplace.

For more information about Access Policy transformations, see the *Transformations* guide in Data Integration.

The following table lists the enforcement environments for each type of data access policy:

Data Access Policy Type	Enforcement Environment
data filter policy	Enforceable in Data Integration and Data Ingestion and Replication through Access Policy transformations and in Data Marketplace when you order a data collection. Enforceable in your source system after policy pushdown from Data Governance and Catalog. You can enforce data filter policies in the following source systems: - Amazon Redshift - Databricks - Microsoft Fabric Data Warehouse - Microsoft Power BI - Snowflake
data de-identification policy	Enforceable in Data Integration and Data Ingestion and Replication through Access Policy transformations and in Data Marketplace when you order a data collection.
data access control policy	Enforceable in your source system after policy pushdown from Data Governance and Catalog. You can enforce data access control policies in the following source systems: - Amazon Redshift - Amazon S3 - Databricks - Google BigQuery - Microsoft Fabric Data lakehouse - Microsoft Fabric Data Warehouse - Microsoft Power BI - Snowflake - Tableau For more information about configuring these source systems for use with data access control policies, see "Prerequisites for pushdown enforcement" on page 17.

Data access policy process

To protect data and control access to your data, you perform steps on the Data Access Management page to create data access policies, rules, and data protections.

To protect and control access to your data, perform the following steps:

1. Determine the data access policy type that best fits your use case.
   For more information, see "Data access policy selection" on page 14.
2. On the Data Access Management page, create a data access policy.
   You can create the following types of data access policies:
   - Data access control policies
   - Data filter policies
   - Data de-identification policies

   For more information, see "Creating data access policies" on page 26.

3. If you determine that you will be creating data de-identification policies, perform the following tasks:
   a. Create precedence tiers so you can select them when you create the policies.
      For more information, see Chapter 3, “Precedence tiers” on page 46.
   b. Create data protections so you can select them when you are create data de-identification rules.
      For more information, see Chapter 4, “Data protections” on page 48.
4. Create one or more rules to add to the data access policy.
   The rule type matches the data access policy type. Rule types include the following rules:
   - Data access control rules
     To create a data access control rule, you perform the following steps:
     1. Select the data access control policy that is associated with the rule.
     2. Give the rule a title and description.
     3. Select the attributes that define the condition for the rule to activate.
     4. Define filters based on data asset types that grant read, write, or delete permissions.

     For more information, see “Creating data access control rules” on page 29.
   - Data filter rules
     To create a data filter rule, you perform the following steps:
     1. Select the data filter policy that is associated with the rule.
     2. Give the rule a title and description.
     3. Select the attributes that define the conditions for the rule to activate.
     4. Define filters based on data classifications that will drop records from the result set.
        For example, if you select a data classification called "Last Name," select the **is any of** operator, and you enter a string value of "Smith," Data Governance and Catalog will drop any record with "Smith" in the "Last Name" column. If instead, you selected the **is not any of** operator, Data Governance and Catalog will keep any record that has the value "Smith" in the "Last Name" column and drop all other records.

     For more information, see “Creating data filter rules” on page 33.
   - Data de-identification rules
     To create a data de-identification rule, you do the following:
     1. Select which data de-identification policy this rule will be associated with.
     2. Give the rule a title and description.
     3. Select the attributes that define the conditions for this rule to activate.
     4. Determine which data classifications to protect and whether they require field-level or cell-level de-identification.
        You only assign one data protection to each data class. You can reuse the same data protection on multiple data classes.
     5. Define field-level de-identifications, if applicable.
        Determine which data protections to use to protect the data classifications. For example, a NAME column might be protected with a tokenized data protection relevant for name data. The data protection applies to the entire column.
     6. Define cell-level de-identifications, if applicable.
        Use cell-level de-identifications when the data that you want to protect is represented in multiple formats. For example, a POSTAL_CODES column might contain both U.S. and Canadian postal codes. Using cell-level de-identifications allows for a different data protection for each type of

postal code. In this case, you can create tokenized data protections that match the format of each type of postal code. Data Governance and Catalog will apply the appropriate data protection based on whether the entry in the COUNTRY column is U.S. or Canada.

7. Optionally, add one or more alternative conditions under which the data protection applies.

For more information, see "Creating data de-identification rules" on page 41.

If you don't have a workflow configured, the data access policy will automatically change to published status.

If you have a workflow configured, the data access policy will change to draft status.

For more information about designing workflows, see the *Administration* help in Metadata Command Center.

Before you begin

Before you can use the features on the Data Access Management page, you assign user permissions, perform catalog tasks, and configure Secure Agent services.

Complete the following tasks:

1. User administration tasks
   In Metadata Command Center, assign one or more users to the Data Access Owner and Stakeholder roles.

   For more information about the Data Access Owner role, see the *Introduction and Getting Started* help in Metadata Command Center.

   You can create custom roles that incorporate the permissions that the Data Access Owner and Stakeholder roles enable.

   For more information about creating a custom role, see the *User Administration* help in Administrator.
2. Data catalog tasks
   a. In Data Marketplace, create terms of use that you can use with data access policy enforcement methods in Data Marketplace, Data Integration, and Data Ingestion and Replication. Terms of use provide usage requirements and guidelines that a user has to accept before they can use data.

      For more information about creating terms of use, see the *Set Up Data Marketplace* help in Data Marketplace.

   b. Optionally, build a glossary and assign terms to technical assets in Data Governance and Catalog.

      For more information about building glossaries, see the *Asset Management* help in Data Governance and Catalog.

   c. In Metadata Command Center, create, configure, and scan catalog sources.

      For more information about creating a catalog source, see the *Catalog Source Configuration* help in Metadata Command Center.

   d. In Data Governance and Catalog, verify that the technical assets from scanned catalog sources have correctly assigned data classifications.

      For more information about scanning technical assets, see the *Introduction and Getting Started* help in Data Governance and Catalog.

3. Data access policy enforcement configuration tasks

   - To enable Data Marketplace to enforce data filter policies and data de-identification policies, configure the Data Access Management Proxy Secure Agent service.

     Note: The Data Access Management Agent and Proxy services will not start without valid values for the `usernameWithConnectionPrivileges` and `userWithConnectionPrivileges` properties.

     Your database vendors supply the source drivers for the Secure Agents. Put these source drivers in the path /ext/connectors/thirdparty/ in all the Secure Agents. For example, if you want to access PostreSQL databases, add the latest PostreSQL driver that you downloaded from a trusted source in the Secure Agent path /ext/connectors/thirdparty/.

     For more information about the Data Access Management Proxy Secure Agent service and the compatible drivers, see the *Secure Agent Services* help in Administrator.

     For more information about creating terms of use, see the "Ordering data collections" help in the Data Marketplace help.

   - To enable your cloud data platform to natively enforce data access control policies, perform the following steps:

     - Configure the Data Access Management Agent Secure Agent service.

       Put a compatible JDBC driver in the following location: <install>/ext/connectors/thirdparty/ where "<install>" represents the directory in which you installed the Secure Agent. You only need to provide these drivers for accessing Snowflake and Databricks cloud data platforms. You can download compatible JDBC drivers from the vendors' websites.

       For more information about the Data Access Management Agent Secure Agent service and the compatible drivers, see the *Secure Agent Services* help in Administrator.

     - Configure your cloud data platform to allow Data Governance and Catalog to apply grants to the technical assets that users specify.

       For more information, see “Prerequisites for pushdown enforcement” on page 17.

     - In Administrator, connect cloud data platform sources.

       For more information about cloud data platform source-specific connections, see *Connections* in the Administrator help.

     - In Metadata Command Center, create and configure your cloud data platform sources. Metadata Command Center scans and classifies the sources during this process.

       For cloud data platform source-specific configuration details, see the *Catalog Source Configuration* help or the *Databricks Sources* help in Metadata Command Center.

     - In Data Governance and Catalog, validate that imported technical assets from scanned catalog sources are in the catalog and that data classifications are correctly assigned for your cloud data platform.

       For more information about importing technical assets, see the *Introduction and Getting Started* help in Data Governance and Catalog.

       Understand how Data Governance and Catalog maps permissions from within data access control policies to cloud data platforms. To learn how Data Governance and Catalog maps permissions, see “Data access control policy permission mappings” on page 24.

4. Configure a workflow event.

   Note: For data access asset types, configure distinct workflow events for asset change and asset creation so that Data Governance and Catalog identifies changes to the Conditions, Rules, Techniques, and De-identification Rank tabs for workflow.

For more information about configuring a workflow event, see *Administration* in the Metadata Command Center help.

5. Create data access policies.

   For more information about creating data access policies, see "Creating data access policies" on page 26.

6. Enforce data access policies.

   - To enforce data filter policies and data de-identification policies in Data Integration, configure a mapping that includes an Access Policy transformation.

     For more information about Access Policy transformations, see the *Transformations* help in Data Integration.

   - To enforce data filter policies and data de-identification policies in Data Ingestion and Replication, configure a replication task with Access Management.

   - To enable Data Marketplace to enforce data filter policies and data de-identification policies, place an order with the **Managed Access** setting turned on in Data Marketplace.

     For more information about placing orders, see the *Set Up Data Marketplace* help in Data Marketplace.

   - To enable your cloud data platform to enforce data access control policies, follow the steps in Data access policy enforcement configuration tasks. Data Governance and Catalog automatically pushes data access control policies to your platform.

     In Metadata Command Center, you can monitor the pushdown of data access control policies to your cloud data platform.

     For more information about monitoring the pushdown of data access control policies in Metadata Command Center, see the *Administration* help in Metadata Command Center.

CHAPTER 2

Data access policies

A data access policy is a set of rules that you can use to protect data and control access to your data. For example, you can write a data access policy around a regulation such as HIPAA or GDPR, or around a business objective such as provisioning data for marketing analytics.

As a Data Steward, you can create data access control policies, data filter policies, and data de-identification policies on the Data Access Management page.

You use data access control policies to grant read, write, or delete access to specific technical assets, including tables, views, and so on in a specified source system.

You use data filter policies to filter out records that Data Consumers should not access due to their role or location.

You use data de-identification policies to remove, alter, or otherwise protect the columns and values in each data asset that the given user can access. Each data de-identification policy contains one or more data de-identification rules, which contain data protections that define which data de-identification techniques to apply to the data based on user and usage context.

Data access policy selection

Before creating data access policies on the Data Access Management page, select which type of policy best fits each of your use cases.

You can create the following types of policies:

- Data access control policies grant groups of users read, write, or delete access to tables or views in a specific source system. Data Governance and Catalog pushes these policies into your cloud data platform. The data source enforces the policy directly.
  Data access control policies best serve the following types of use cases:
  - Provisioning data for analytics in native cloud data platforms using native access controls and tools where some groups only require read access to entire tables or views of data while other groups require write access
- Data filter policies restrict or limit which records Data Governance and Catalog delivers to users.
  Data filter policies best serve the following types of use cases:
  - Complying with data privacy regulations such as HIPAA, GDPR, or data transfer regulations by limiting which records users can access
  - Provisioning data for analytics in which the analysts have only access to some of the data

- Data de-identification policies transform data in ways that you specify to de-identify sensitive data. The de-identified data might be tokenized to maintain the original format, generalized to a common year or month, or truncated to keep only minimal contents visible, making it useful for analysis.
  Data de-identification policies best serve the following types of use cases:
  - Complying with data privacy regulations such as HIPAA or GDPR while still being able to use the de-identified data for analysis
  - Provisioning data for analytics in which the analysts must not have access to raw, un-redacted data

Example scenarios

The following are examples that illustrate which type of data access policy to use in different business scenarios.

The Operations team at your financial organization needs write access to any customer data in accordance with financial regulations. Your organization's anti-money laundering (AML) department needs read access to customer data to ensure that no customers are laundering money through your organization. All of your customer data is stored in a Snowflake source system. Data access control policies best serve this use case because you can write policies that Data Governance and Catalog pushes directly into your Snowflake instance. For tables containing customer data, the data access policies can grant write access to the Operations team while granting read access to the AML department.

You have customers in Western Europe and the Middle East. Your organization stores customer data in data centers in Dublin, Ireland and Riyadh, Saudi Arabia. Data filter policies best serve this use case because you can write policies that ensure that European customer data is only accessed and processed within the European Union and Saudi Arabian customer data is only accessed and processed within Saudi Arabia. This prevents your organization from running afoul of data sovereignty regulations included in the GDPR and the PDPL.

The compliance department at your healthcare organization wants to perform analytics on patient data to determine what percentage of patients are overdue for more than one preventative health screening. The patient data includes patient ID numbers, contact information, and medical history. Data de-identification policies best serve this use case because you can write policies that replace patient ID numbers with tokens, redact all contact information, and leave medical history untouched. Using the tokenized patient ID numbers, your compliance department can determine how many preventative health screenings each patient is overdue for without being able to identify any individual patient. This prevents your organization from running afoul of data privacy regulations such as HIPAA.

Guidelines for sequencing data access policies for application

As a Data Steward, you manage data access policies to determine under what conditions Data Governance and Catalog processes them.

The following list describes guidelines for applying each type of policy:

- You don't need to determine the order of data access control policies because they are cumulative. If a policy grants a user group read access to a given table, and another policy grants that user group write access to the same table, the user group will have read and write access to that table.
- You don't need to determine the order of data filter policies because they are cumulative. For each policy that applies in a given context, Data Governance and Catalog increasingly restricts access to the data by filtering the rows returned. Data Governance and Catalog processes all data filter policies simultaneously,

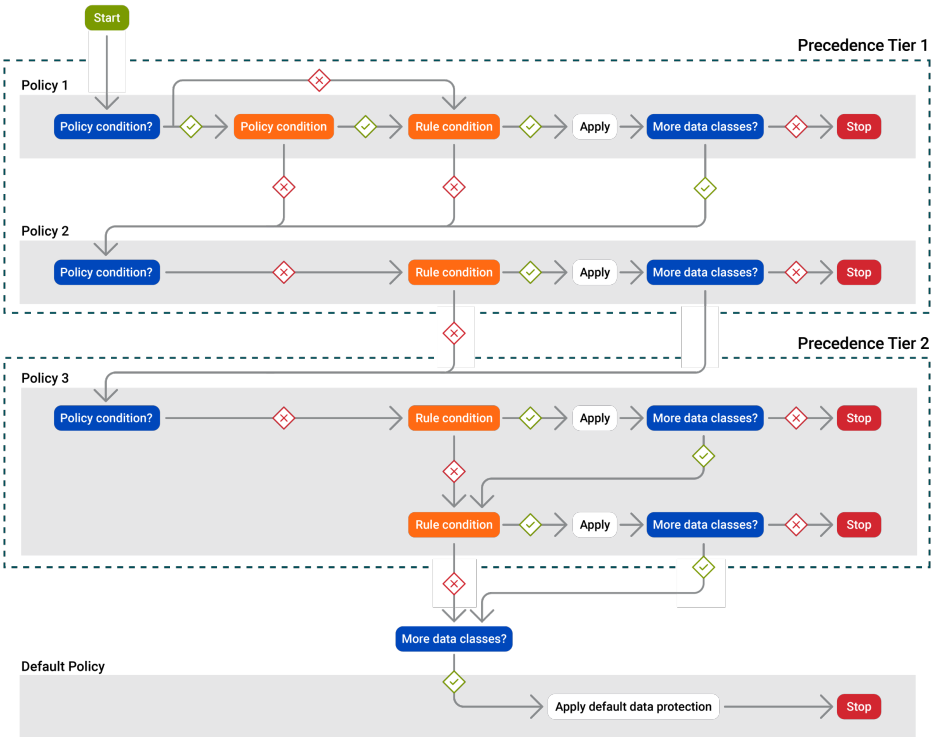
and within each policy, Data Governance and Catalog processes all data filter rules simultaneously. If multiple data filter rules match the context, Data Governance and Catalog applies all those rules.

- You need to determine the order of data de-identification policies.
  - You place each data de-identification policy in a precedence tier. Data Governance and Catalog processes precedence tiers based on rank from lowest integer to highest.
  - Within each precedence tier, Data Governance and Catalog processes data de-identification policies from top to bottom.
  - Within each data de-identification policy, Data Governance and Catalog processes data de-identification rules from top to bottom. If multiple data de-identification rules match the set of conditions for a given request, Data Governance and Catalog runs the first rule in order, applies data protections to any relevant fields based on data element classifications, and then prevents any other rules from executing on the same field.
  - Within data de-identification rules, Data Governance and Catalog processes the sequence of classifications within the rule from top to bottom. Cell-level de-identifications take precedence over field-level de-identifications.

When Data Governance and Catalog processes data de-identification policies, it starts by processing a precedence tier. Inside the precedence tier are policies. Data Governance and Catalog evaluates the first policy. It first determines whether there is a policy condition. If there is, it evaluates that condition. Since that condition applies, it evaluates the rule condition. If that condition applies, it applies the data protection to the data associated with that data class. If there are more data classes, the process repeats until there are either no more data classes to evaluate or no conditions that apply. Data Governance and Catalog applies the default data protection to any remaining data classes.

The following image shows the data de-identification policy evaluation process:

How Data Access Management processes data protection policies



Prerequisites for pushdown enforcement

To enable Data Governance and Catalog to push down data access control policies and data filter policies into your source system, complete the necessary configuration and authorization tasks.

The following table lists the types of data access policies that you can push down in each type of source system:

Source System Type	Data Access Policy Types
Amazon Redshift	Data access control policies and data filter policies
Amazon S3	Data access control policies
Databricks	Data access control policies and data filter policies
Google BigQuery	Data access control policies
Microsoft Fabric Data Lakehouse	Data access control policies and data filter policies
Microsoft Fabric Data Warehouse	Data access control policies and data filter policies
Microsoft Power BI	Data access control policies and data filter policies
Snowflake	Data access control policies and data filter policies
Tableau	Data access control policies

After you configure your source system to work with pushdown enforcement, you assign permissions to data assets.

For more information, see "Assigning permissions to source systems" on page 31.

Prerequisites for Amazon Redshift pushdown enforcement

You can enable Data Governance and Catalog to push down data access control policies and data filter policies into your Amazon Redshift source system.

Complete the following configuration and authorization tasks for your Amazon Redshift source system:

1. Configure the connection.
   For more information about creating an Amazon Redshift connection, see the *Amazon Redshift Sources* guide in Metadata Command Center.
2. Configure Amazon Redshift as a catalog source.
   For more information about configuring a catalog source for Amazon Redshift, see the *Catalog Source Configuration* help in Metadata Command Center.
3. Grant the following privileges to the connection user associated with your Amazon Redshift source system:

```
grant create role to role [IDMC_USER_ROLE];
grant { { SELECT | INSERT | UPDATE | DELETE } [,...] | ALL [ PRIVILEGES ] } on
[OBJECT_NAME]
to { [IDMC_USER_ROLE] with grant option;
```

To grant privileges on an object in Amazon Redshift, meet one of the following criteria::

- Be the object owner.
- Be a superuser.
- Have a grant privilege for that object and for the privilege that you'll grant.

For more information about configuring connection properties for Amazon Redshift, see the *Connections* help in Administrator.

Note: If your organization uses an identity provider (IdP) and pushes data access policies to Amazon Redshift, add a custom property to the Data Access Management Agent service for the namespace that Amazon Redshift requires. The custom property allows the Secure Agent to map the IDMC user groups in the data access policies into the IdP-based roles created in a namespace in Amazon Redshift.

For more information about adding properties to the Data Access Management Agent service, see the *Secure Agent Services* help in Administrator.

Prerequisites for Amazon S3 pushdown enforcement

You can enable Data Governance and Catalog to push down data access control policies into your Amazon S3 source system.

Data access control policies use AWS customer-managed policies to determine access to data. The policies must be attached to an IAM role.

Complete the following configuration and authorization tasks for your Amazon S3 source system:

1. Configure the connection.
   For more information about creating an Amazon S3 connection, see the *Amazon S3 Sources* guide in Metadata Command Center.
2. Configure Amazon S3 as a catalog source.
   For more information about configuring a catalog source for Amazon S3, see the *Catalog Source Configuration* help in Metadata Command Center.
3. Grant the following privileges to the connection user associated with your Amazon S3 source system:
   - `sts:GetCallerIdentity`. Enables resolution of the AWS account ID during client initialization.
   - `iam:GetRole`. Ensures the target IAM role exists before attaching IAM policies.
   - `iam:CreatePolicy`. Allows creation of new managed IAM policies such as S3 permissions.
   - `iam:GetPolicy`. Verifies that an IAM policy exists and to retrieve its metadata.
   - `iam:DeletePolicy`. Removes managed policies when access is revoked.
   - `iam:ListPolicies`. Enables the listing of AWS customer-managed policies.
   - `iam:CreatePolicyVersion`. Updates an IAM policy by creating a new version.
   - `iam:GetPolicyVersion`. Retreives the IAM policy document content from a specified version.
   - `iam:ListPolicyVersion`. Lists all versions of an IAM policy. This is useful for cleanup and version management.
   - `iam:DeletePolicyVersion`. Deletes policy versions.

     Note: Amazon S3 enforces a maximum of five versions for each IAM policy.
   - `iam:AttachRolePolicy`. Attaches AWS customer-managed IAM policies to IAM roles to grant permissions.
   - `iam:DetachRolePolicy`. Deattaches AWS customer-managed IAM policies from IAM roles to revoke permissions.

For more information about configuring connection properties for Amazon S3, see the *Connections* help in Administrator.

Prerequisites for Databricks pushdown enforcement

You can enable Data Governance and Catalog to push down data access control policies and data filter policies into your Databricks source system.

Complete the following configuration and authorization tasks for your Databricks source system:

1. Configure the connection.
   For more information about creating a Databricks connection, see the *Databricks Sources* guide in Metadata Command Center.
2. Ensure that the user identified in the catalog source connection that pushes the policies has Databricks workspace admin permissions on the catalog source.
3. Configure Databricks as a catalog source.
   For each catalog and schema on which you want to apply data access policies, there are two options for granting permissions:
   - Grant MANAGE and USAGE on the catalog and USAGE on all relevant schemas:

     ```
     GRANT USAGE ON CATALOG catalog_name TO user_or_role;
     GRANT MANAGE ON CATALOG catalog_name TO user_or_role;
     GRANT USAGE ON SCHEMA catalog_name.schema TO user_or_role;
     ```

   - Grant USAGE on the catalog and USAGE and MANAGE on all relevant schemas:

     ```
     GRANT USAGE ON CATALOG catalog_name TO user_or_role;
     GRANT USAGE ON SCHEMA catalog_name.schema TO user_or_role;
     GRANT MANAGE ON SCHEMA catalog_name.schema TO user_or_role;
     ```

4. For use with data filter policies, your Databricks catalog needs to be able to enforce Databricks' row filters.
   Data Access Management requires a Databricks catalog to store the necessary objects and functions.

   Data Access Management uses the following Databricks catalog by default:

   ```
   cdam_internal_state
   ```

5. For data filter policies, create the following catalog and schema on your Databricks source system:

   ```
   CREATE cdam_internal_state.default
   ```

   Note: Databricks does not allow row filters on views. Therefore, you can't apply data filter policies to views in Databricks.
6. For data filter policies, grant permissions to the connection associated with your Databricks source system:

   ```
   GRANT CREATE FUNCTION ON SCHEMA cdam_internal_state.default TO user_or_role;
   GRANT DROP FUNCTION ON SCHEMA cdam_internal_state.default TO user_or_role;
   ```

For more information about configuring a catalog source for Databricks, see the *Databricks Sources* help in Metadata Command Center.

Prerequisites for Google BigQuery pushdown enforcement

You can enable Data Governance and Catalog to push down data access control policies into your Google BigQuery source system.

Complete the following configuration and authorization tasks for your Google BigQuery source system:

1. Configure the connection.
   For more information about creating a Google BigQuery connection, see the *Google BigQuery Sources* guide in Metadata Command Center.
2. Configure Google BigQuery as a catalog source.
   For more information about configuring a catalog source for Google BigQuery, see the *Catalog Source Configuration* help in Metadata Command Center.
3. In Google BigQuery, create a custom role with the following permissions per project:
   - `roles/bigquery.user`. Runs statements on Google BigQuery.
   - `bigquery.tables.setIamPolicy`. Grants or revokes roles for assets.
   - `iam.roles.get`. Views role state information.
   - `iam.roles.undelete`. Undeletes deleted custom roles.
   - `iam.roles.update`. Switches to disabled state.
   - `iam.roles.create`. Creates custom roles.
   - `iam.roles.list`. Views role information.
   - Optional: If you don't want to give `iam.roles.create` and `iam.roles.list` permissions to Data Governance and Catalog, you can instead create the following custom roles in each project:
     - CDAM read role
       - roleId: CDAMRead
       - permission: bigquery.tables.getData
     - CDAM write role
       - roleId: CDAMWrite
       - permission: bigquery.tables.updateData

   Alternatively, grant the following roles to the Google BigQuery service account that the connection uses:
   - `roles/bigquery.dataOwner`. Grants or revokes roles.
   - `roles/iam.roleAdmin`. Creates custom roles.

For more information about configuring connection properties for Google BigQuery, see the *Connections* help in Administrator.

Prerequisites for Microsoft Fabric Data Lakehouse pushdown enforcement

You can enable Data Governance and Catalog to push down data access control policies and data filter policies into your Microsoft Fabric Data Lakehouse source system.

Complete the following configuration and authorization tasks for your Microsoft Fabric Data Lakehouse source system:

1. Configure the connection.
   For more information about creating a Microsoft Fabric Data Lakehouse connection, see the *Microsoft Fabric Data Lakehouse Sources* guide in Metadata Command Center.

2. Configure Microsoft Fabric Data Lakehouse as a catalog source.
   For more information about configuring a catalog source for Microsoft Fabric Data Lakehouse, see the *Connections* help in Administrator.
3. For each database into which you will push data access control policies and data filter policies, grant the service principal the following permissions:
   - CREATE ROLE

     ```
     GRANT CREATE ROLE ON DATABASE::[DATABASE_NAME] TO [IDMC_USER_ROLE];
     GO
     ```

   - ALTER ANY ROLE

     ```
     GRANT ALTER ANY ROLE ON DATABASE::[DATABASE_NAME] TO [IDMC_USER_ROLE];
     GO
     ```

   - CONTROL

     ```
     GRANT CONTROL ON DATABASE::[DATABASE_NAME] TO [IDMC_USER_ROLE];
     GO
     ```

   Alternatively, grant the following permission to users who need to grant any permission on any database in the server:

   - CONTROL SERVER

     ```
     GRANT CONTROL SERVER TO [IDMC_USER_ROLE];
     GO
     ```

4. For data filter policies, additionally grant the following permission on the database:

   ```
   GRANT CREATE SCHEMA TO [IDMC_USER_ROLE]
   GRANT CREATE FUNCTION TO [IDMC_USER_ROLE]
   ```

   Note: The default Microsoft Fabric Data Warehouse schema name that Data Access Management uses to manage data filter policies is CDAM_INTERNAL_STATE. If this name does not comply with your organization's schema naming convention, use the plugin.fabric-warehouse.default.schema property to give the schema another name.

   Note: If you manually created the CDAM_INTERNAL_STATE schema, add the following permission for data filter policies:

   ```
   GRANT ALTER ON SCHEMA::CDAM_INTERNAL_STATE TO [IDMC_USER_ROLE]
   ```

5. For data filter policies, additionally grant permissions to create a security policy for row-level security on the schema in the following format:

   ```
   GRANT ALTER ANY SECURITY POLICY TO [IDMC_USER_ROLE]
   ```

Prerequisites for Microsoft Fabric Data Warehouse pushdown enforcement

You can enable Data Governance and Catalog to push down data access control policies and data filter policies into your Microsoft Fabric Data Warehouse source system.

Complete the following configuration and authorization tasks for your Microsoft Fabric Data Warehouse source system:

1. Configure the connection.
   For more information about creating a Microsoft Azure Fabric connection, see the *Microsoft Fabric Data Warehouse Sources* guide in Metadata Command Center.
2. Configure Microsoft Fabric Data Warehouse as a catalog source.
   For more information about configuring a catalog source for Microsoft Fabric Data Warehouse, see the *Connections* help in Administrator.

3. For each database into which you will push data access control policies and data filter policies, grant the service principal the following permissions:
   - CREATE ROLE

     ```
     GRANT CREATE ROLE ON DATABASE::[DATABASE_NAME] TO [IDMC_USER_ROLE];
     GO
     ```

   - ALTER ANY ROLE

     ```
     GRANT ALTER ANY ROLE ON DATABASE::[DATABASE_NAME] TO [IDMC_USER_ROLE];
     GO
     ```

   - CONTROL

     ```
     GRANT CONTROL ON DATABASE::[DATABASE_NAME] TO [IDMC_USER_ROLE];
     GO
     ```

   Alternatively, grant the following permission to users who need to grant any permission on any database in the server:
   - CONTROL SERVER

     ```
     GRANT CONTROL SERVER TO [IDMC_USER_ROLE];
     GO
     ```

4. For data filter policies, additionally grant the following permission on the database:

   ```
   GRANT CREATE SCHEMA TO [IDMC_USER_ROLE]
   GRANT CREATE FUNCTION TO [IDMC_USER_ROLE]
   ```

   Note: The default Microsoft Fabric Data Warehouse schema name that Data Access Management uses to manage data filter policies is CDAM_INTERNAL_STATE. If this name does not comply with your organization's schema naming convention, use the plugin.fabric-warehouse.default.schema property to give the schema another name.

   Note: If you manually created the CDAM_INTERNAL_STATE schema, add the following permission for data filter policies:

   ```
   GRANT ALTER ON SCHEMA::CDAM_INTERNAL_STATE TO [IDMC_USER_ROLE]
   ```

5. For data filter policies, additionally grant permissions to create a security policy for row-level security on the schema in the following format:

   ```
   GRANT ALTER ANY SECURITY POLICY TO [IDMC_USER_ROLE]
   ```

Prerequisites for Microsoft Power BI pushdown enforcement

You can enable Data Governance and Catalog to push down data access control policies and data filter policies into your Microsoft Power BI source system.

Complete the following configuration and authorization tasks:

1. Use one of the following types of Microsoft workspaces:
   - Fabric
   - Power BI Premium
   - Power BI Premium Per User
2. Configure the connection.
   For more information about creating a Microsoft Power BI connection, see the *Microsoft Power BI Sources* guide in Metadata Command Center.
3. In Metadata Command Center configure Microsoft Power BI as a catalog source.
   For more information about configuring a catalog source for Microsoft Power BI, see the *Connections* help in Administrator.
4. Disable the tenant-level setting "Block republish and disable package refresh" to prevent access issues related to Microsoft Power BI permissioning.

5. For each Microsoft Power BI workspace into which you want Data Governance and Catalog to push data access control policies and data filter policies, add the user or service principal as a member or admin of the workspace.
6. Ensure that the XMLA endpoint property is set to read-write.
7. Grant the following permissions as the Delegated type to the connection user associated with your Microsoft Power BI source system:

```
Dataset.ReadWrite.All
Dataset.Read.All
Workspace.ReadWrite.All
```

8. Grant the following permissions as the Application type to the connection user associated with your Microsoft Power BI source system:

```
Group.Read.All
GroupMember.Read.All
```

Note: Grant administrator consent in Microsoft Power BI to approve application permissions.

Prerequisites for Snowflake pushdown enforcement

You can enable Data Governance and Catalog to push down data access control policies and data filter policies into your Snowflake source system.

Complete the following configuration and authorization tasks for your Snowflake source system:

1. Configure the connection.
   For more information about creating a Snowflake connection, see the *Snowflake Sources* guide in Metadata Command Center.
2. Configure Snowflake as a catalog source.
   For more information about configuring a catalog source for Snowflake, see the *Catalog Source Configuration* help in Metadata Command Center.
3. Determine which types of data access policies you would like to enforce in your Snowflake source system. You can enforce data access control policies and data filter policies. Each requires different permissions.
4. For use with data access control policies, grant the following permissions to the user role associated with your Snowflake source system connection:

```
GRANT MANAGE GRANTS ON ACCOUNT TO [IDMC_USER_ROLE];
GRANT CREATE ROLE ON ACCOUNT TO [IDMC_USER_ROLE];
```

5. For use with data filter policies, your Snowflake account needs to be able to enforce Snowflake's row access policies.
   To enforce row access policies, Data Access Management requires a Snowflake database to store the necessary objects. You can configure this database in any of the following ways:
   - The Data Access Management Secure Agent service can create it automatically, if you grant it the following permission:

     ```
     GRANT CREATE DATABASE ON ACCOUNT TO [IDMC_USER_ROLE];
     ```

   - You can create the database yourself by creating a database with the name `CDAM_INTERNAL_STATE` and granting at a minimum the following permissions to the Snowflake role that you associate with the connection you create in IDMC:

     ```
     GRANT USAGE ON CDAM_INTERNAL_STATE TO [IDMC_USER_ROLE];
     GRANT CREATE ROW ACCESS POLICY ON SCHEMA "CDAM_INTERNAL_STATE"."PUBLIC" TO ROLE
     [IDMC_USER_ROLE];
     ```

For all databases for which you want to apply row access policies, grant the following permission:

```
GRANT USAGE ON DATABASE [DATABASE_NAME] TO ROLE [IDMC_USER_ROLE];
```

Regardless of who created the database, grant the following permission to the user in the connection:

```
GRANT APPLY ROW ACCESS POLICY ON ACCOUNT TO [IDMC_USER_ROLE];
```

Prerequisites for Tableau pushdown enforcement

You can enable Data Governance and Catalog to push down data access control policies into your Tableau source system.

Complete the following configuration and authorization tasks for your Tableau source system:

1. Configure the connection.
   For more information about creating a Tableau connection, see the *Tableau Sources* guide in Metadata Command Center.
2. Configure Tableau as a catalog source.
   For more information about configuring a catalog source for Tableau, see the *Catalog Source Configuration* help in Metadata Command Center.
3. Configure the user account with the following permissions:
   - Interactor license level
   - View and download permissions for all projects, workbooks, and catalog sources that you want Data Governance and Catalog to act upon

   Note: Tableau automatically grants permissions from parent objects to their child objects such as tables within a database.

   For more information about configuring connection properties for Tableau, see the *Connections* help in Administrator.

Data access control policy permission mappings

When you configure a data access control policy to grant data access permissions to a user in a source system, bear in mind that different platforms apply the policy permissions in different ways.

The following table lists the permissions that you can configure in a data access control policy and the corresponding permissions that the policy enables in source systems:

Source System	Read Equivalent	Write Equivalent	Delete Equivalent
Amazon Redshift	select	insert update	delete
Amazon S3	s3:GetObject s3:ListBucket	s3:PutObject s3:RestoreObject s3:AbortMultipartUpload s3:ListMultipartUploadParts	s3:DeleteObject

Source System	Read Equivalent	Write Equivalent	Delete Equivalent
Databricks	select	modify	(not applicable)
Google BigQuery	bigquery.tables.getData	bigquery.tables.updateData (grants insert, update, and delete)	bigquery.tables.updateData (grants insert, update, and delete)
Microsoft Fabric Data Lakehouse	select	insert update	delete
Microsoft Fabric Data Warehouse	select	insert update	delete
Microsoft Power BI	For Power BI workspaces: Viewer role For Power BI data sets: Viewer role	For Power BI workspaces: Contributor role Not applicable for Power BI data sets	(not applicable)
Snowflake	select	insert update	delete
Tableau	view template	For Tableau projects: publish template For Tableau workbooks: download, edit, overwrite, save as For Tableau data sources: download, overwrite, save as For Tableau data sheets: download, edit	delete (not applicable for Tableau projects)

Configuration guidelines

Consider the following guidelines when you configure data access control policy permissions:

General guidelines

Consider the following general guidelines when you configure data access control policy permissions:

- Because views are read-only objects, a source system ignores permissions other than read when a policy applies to a view.
- The delete permission doesn't apply to Databricks or Microsoft Power BI. If you select the delete permission, you don't grant any permission.

Databricks

Consider the following guidelines when you configure data access control policy permissions for Databricks:

- The Databricks modify permission grants write and delete access. If you grant write permission to a Databricks object, you also implicitly grant the delete permission.
- For the Databricks Unity and Hive catalog types, Data Governance and Catalog grants user permissions to catalogs and schemas.

- The Data Access Management Agent service skips any sample tables and system tables when trying to enforce data access control policies. This is because you can't manage sample tables and system tables unless you use an account admin in the connection.

Google BigQuery

Consider the following guidelines when you configure data access control policy permissions for Google BigQuery:

- Users who read data in views need the select permission on their sources.
- Users need select permission in addition to insert, update, and delete.

Microsoft Power BI

Consider the following guidelines when you configure data access control policy permissions for Microsoft Power BI:

- If a data access control policy grants read access to a data set, users have access to all tables in the data set.
- Data filter policies apply to users who have the Viewer role. Any users with other Power BI roles will see all data.
- The Contributor role grants write and delete access to workspace objects. If you grant write permission to a Microsoft Power BI workspace object, you also implicitly grant the delete permission.

Microsoft Fabric Data Warehouse

Users need read access in addition to delete access to delete a table or view.

Snowflake

For Snowflake catalogs, Data Governance and Catalog grants user permissions to databases and schemas.

Creating data access policies

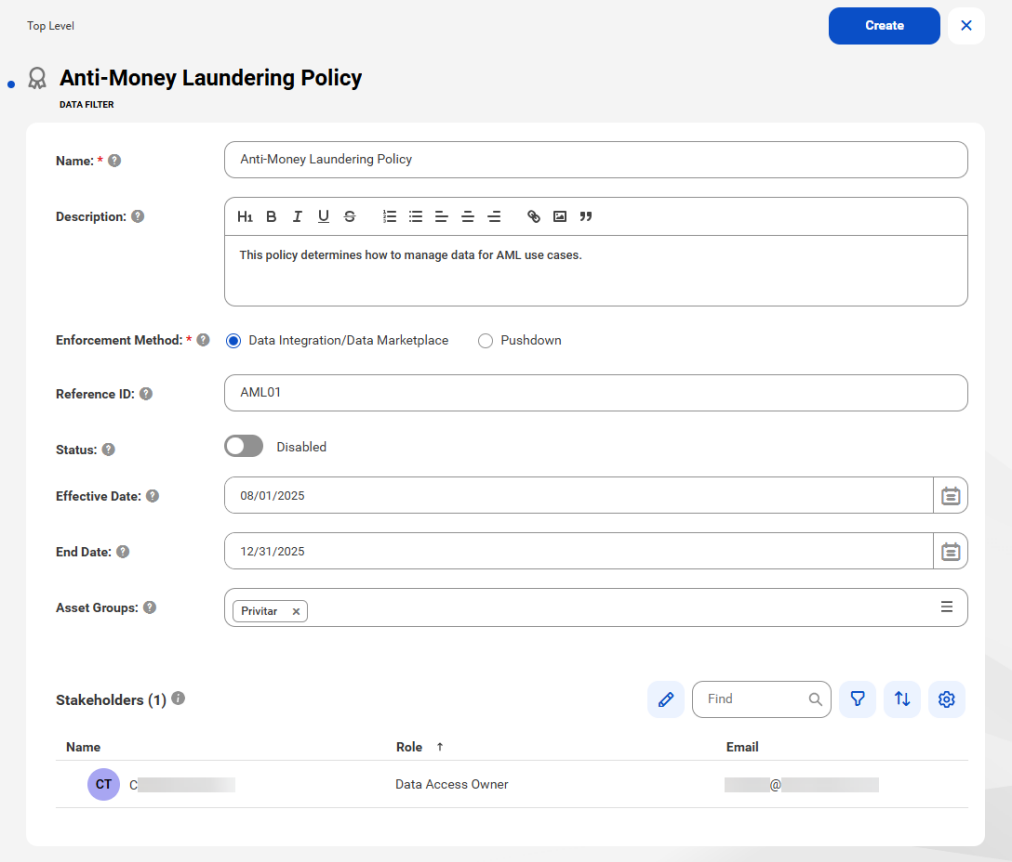
You create data access policies to control access to tables or views in cloud data platforms and to filter or de-identify data in other data assets. The steps for creating data access policies are mostly the same, regardless of which type of data access policy you create.

1. Click **Data Access** in the left navigation to display the Data Access Management page.
2. Select a type of data access policy by clicking the corresponding tab.

   You can create the following types of data access policies:

   - Data access control policies
   - Data filter policies
   - Data de-identification policies
3. Click the plus sign to open the data access policy creation page.

The following image shows the data access policy creation page for a data filter policy:



4. Enter the following properties for the data access policy:

Field	Description
Name	Name of the data access policy.
Description	Description of the data access policy.
Enforcement Method	The enforcement method for the data access policy. - Select **Data Integration/Data Marketplace** to have Data Access Management use this data access policy to protect data in Data Integration mappings and Data Marketplace orders. - Select **Pushdown** to have Data Access Management push down the data access policy into your cloud data platform. **Note:** Not all enforcement methods are available for all types of data access policies.
Precedence Tier	The precedence tier of which this data access policy is a part. You organize data de-identification policies in precedence tiers. You don't organize data access control policies and data filter policies in precedence tiers.
Reference ID	Unique identifier for the data access policy.

Field	Description
Status	The status of the data access policy. Enable the data access policy if you want it to be active immediately after you publish it.
Effective Date	The effective date of the data access policy.
End Date	The end date of the data access policy.
Asset Groups	The asset groups to which the data access policy is assigned.
Stakeholders	The stakeholders for the data access policy. **Note:** Add stakeholders to ensure that you have access to this data access policy when you configure a workflow.

5. Click **Create**.

   The overview page for the data access policy appears.

You can now add rules to this policy.

If you don't have a workflow configured, this data access policy automatically changes to published status.

If you do have a workflow configured, this data access policy changes to draft status.

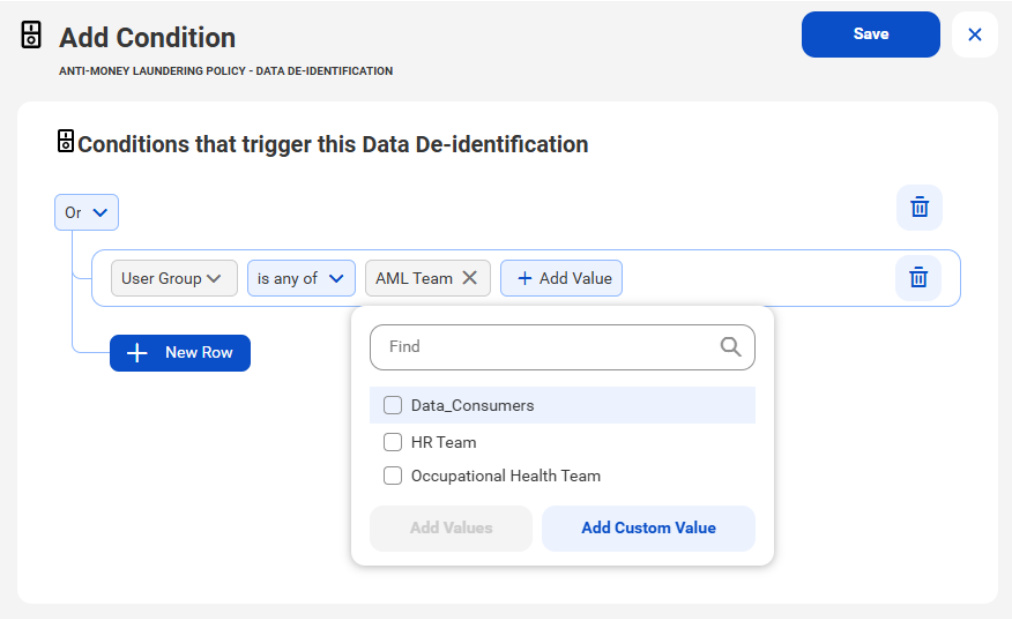
For more information about designing workflows, see the *Administration* help in Metadata Command Center.

Adding a condition to a data access policy

After you create a new data access policy, you can add one or more conditions to it to further refine when the policy will be activated. This is optional. Only data de-identification policies and data filter policies with an enforcement method of "Data Integration/Data Marketplace" have conditions. The steps for adding a condition to a data access policy are mostly the same, regardless of the type of data access policy.

1. View a data access policy.
2. Click the **Conditions** tab.
3. Click the plus sign.

The **Add Condition** page appears. The following image shows the **Add Condition** page:



4. Click **New Row**.
5. Select a contextual attribute, such as **User Group** or **User Context**.
6. Select an operator, such as **is any of** or **is null**.

   Note: Don't create a data access policy with the User Group condition "is null" because it is unlikely to ever apply.
7. Specify a value.
8. Add additional values as needed.
9. Click **Save**.

   You can add rules to this data access policy. The type of rules that you add depends on the type of data access policy.

Creating data access control rules

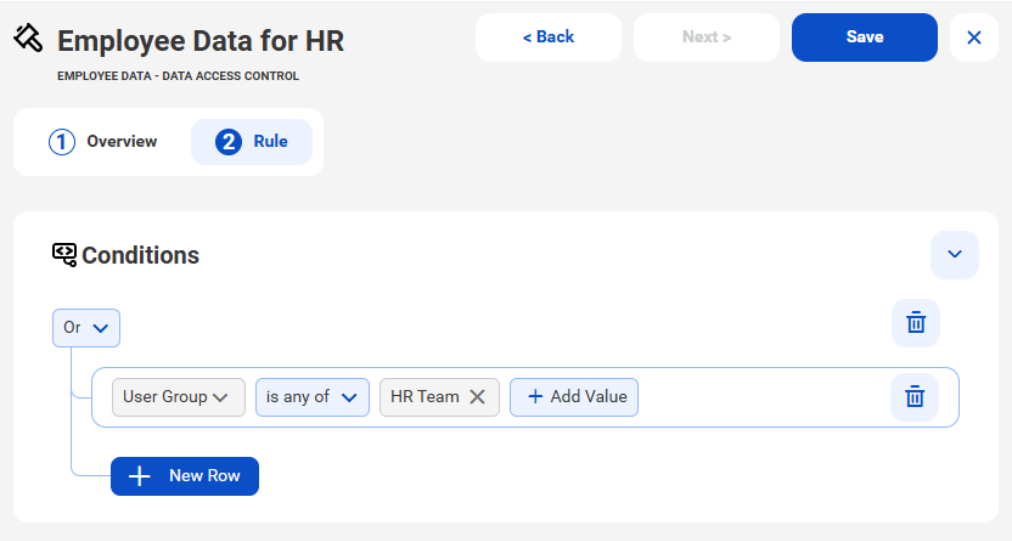
You create data access control rules to grant read, write, or delete access to assets in a cloud data platform, also called a source system for a set of user groups that you specify.

1. Open a data access control policy and select the **Rules** tab.
2. Click the plus sign.

   The **Rule Overview** page appears.
3. Enter a name and description for the rule.
4. Click **Next**.

   The **Rule** tab appears. On this tab, you specify the conditions under which Data Access Management will grant read, write, or delete access to assets in a source system.

The following image shows the **Rule** tab:



5. In the **Conditions** section, select a user group, an operator, and a value.
6. You can choose from the list of known user groups in IDMC if those are the same user groups used in your cloud data platform, or you can add a user group as a custom value. If you add a custom value, enter the name of the role or the account group to which the permissions will be granted.

   Note: Some catalog sources like Snowflake have case-sensitive role names. If the rule references Snowflake and another cloud data platform's assets, treat the role name or group name as case-sensitive.
7. Optionally, click the plus sign again to add another user group.

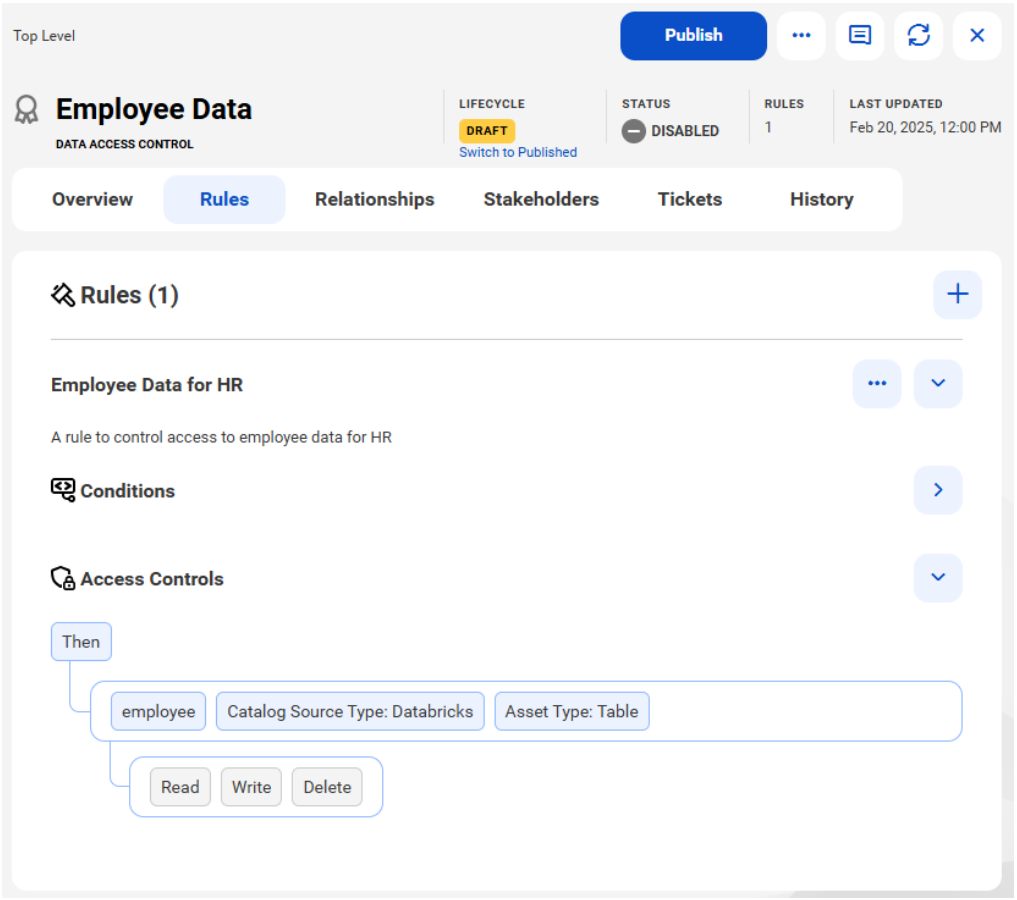
   You can now add access control filters.

Assigning permissions to source systems

After you create conditions in a data access control rule, you then assign permissions to data assets by creating access control filters. You then save the data access control rule to data access control policies.

1. In the **Access Controls** section, Click **New Row**.

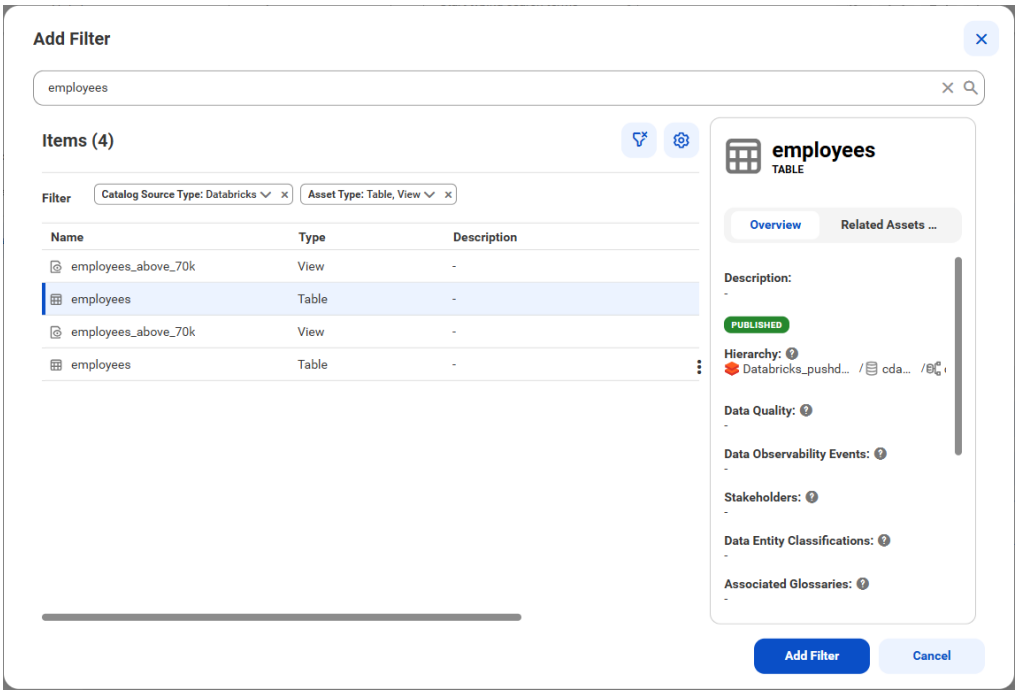
   The following image shows the **Access Controls** section of the **Rule** tab:



2. Click **Add Filter** to select one or more catalog source types and data assets.

   The **Add Filter** window appears.

The following image shows the **Add Filter** window:



3. Enter a search query to find the data assets for which you want to grant permissions.

   The **Preview** section shows the current results of the query. The **Asset Type** and **Catalog Source Type** filters are pre-populated. Adjust these filters and your search query to refine the results as needed.

   Note: Data access control policies only work with specific catalog source types.

   For more information on catalog source types, see *Catalog Source Quick Reference* in the Data Governance and Catalog help.

   Note the following about query results:

   - The preview returns results based on currently accessible metadata. This might change with the addition of other assets that match the filter query.
   - Don't use the universal wildcard * because it grants access to all assets within the identified Data Governance and Catalog catalog source type.
   - A single permission assignment has a limit of 2,000 assets. If your query results in more than 2,000 assets, you can partition your filter across multiple rules to reduce the number of assets in the query.
4. Click **Add Filter** to return to the **Access Controls** section of the **Rule** tab.
5. Click **New Permission**.
6. Select any combination of **Read**, **Write**, and **Delete**. **Read** is selected by default.

   Note: Because views are read-only objects, only read permissions will apply even if your rule offers write and delete permissions.

7. Save the rule.

   Your updates take effect when the policy associated with the rule is published.
   If you don't have a workflow configured, this rule will change to published status.

   If you have a workflow configured, this rule and its associated policy will change to draft status.

   For more information about designing workflows, see the *Administration* help in Metadata Command Center.

   Note: You can monitor the process to push down the policy initiated by using the data access sync job type on the **Monitor** page in Metadata Command Center.

   For more information about data access sync jobs, see the *Administration* help in Metadata Command Center.

Data filter policy behavior

Data filter policies are additive and deny access to rows of data.

Data filter policies function in the following ways:

- If there are no data filter policies, Data Governance and Catalog grants access to all rows of data.
- For each data filter rule, if the condition is met, Data Governance and Catalog denies access to the rows.
- If a data element has multiple data classifications, Data Governance and Catalog denies access to the row if any of the classifications matches the filter criteria.
- Data filter policies are additive. If one policy denies access to one row, and a second policy denies access to a second row, the user doesn't have access to either row if both policies apply.

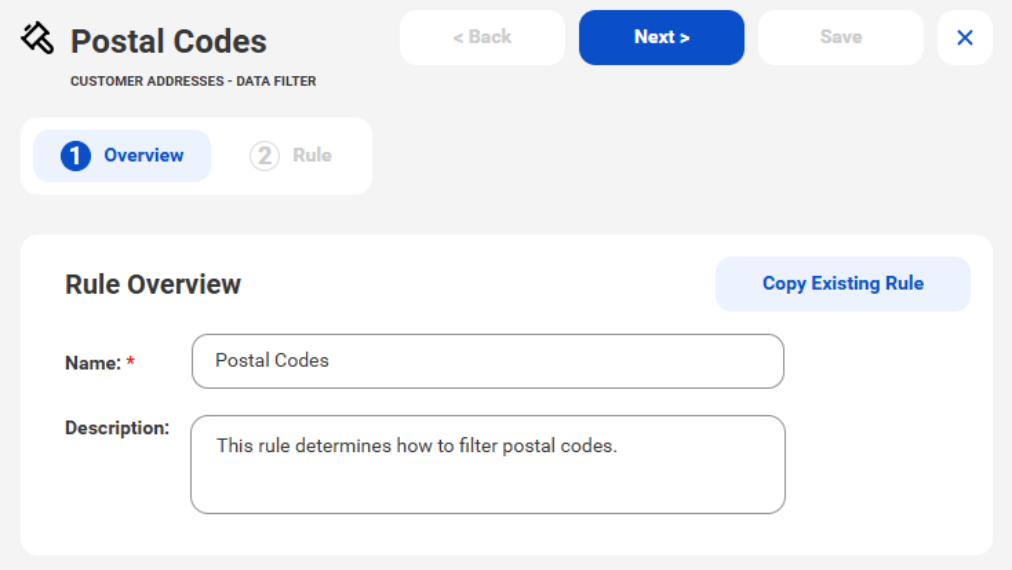
Creating data filter rules

You create data filter rules to control access to rows or records of data.

1. Open a data filter policy and select the **Rules** tab.
2. Click the plus sign.

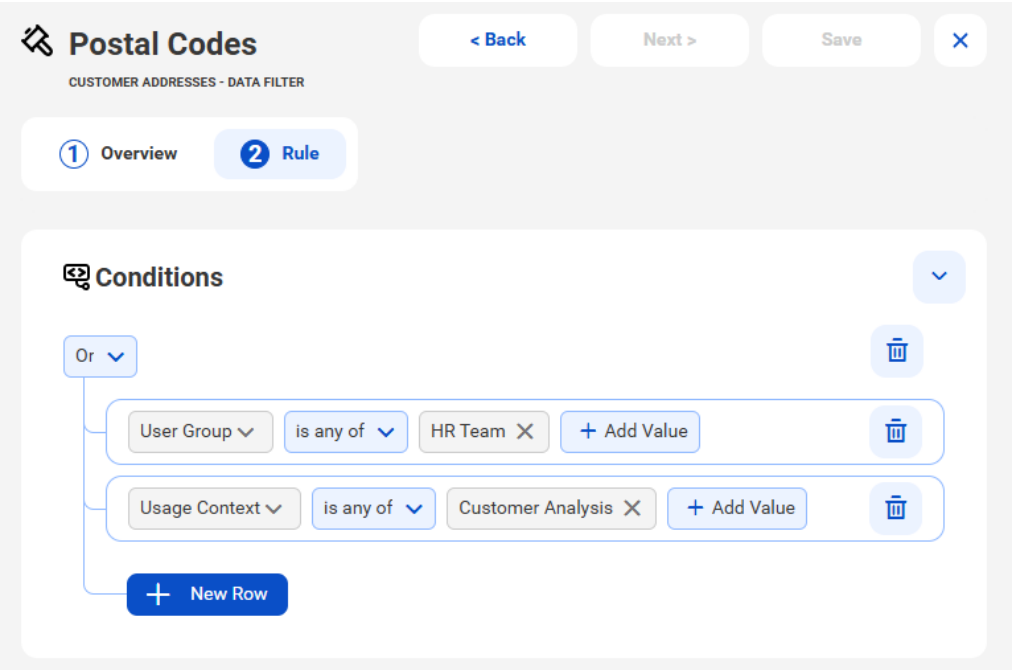
   The **Overview** tab appears.
3. Enter a name and a description for the rule.

The following image shows the **Overview** tab:

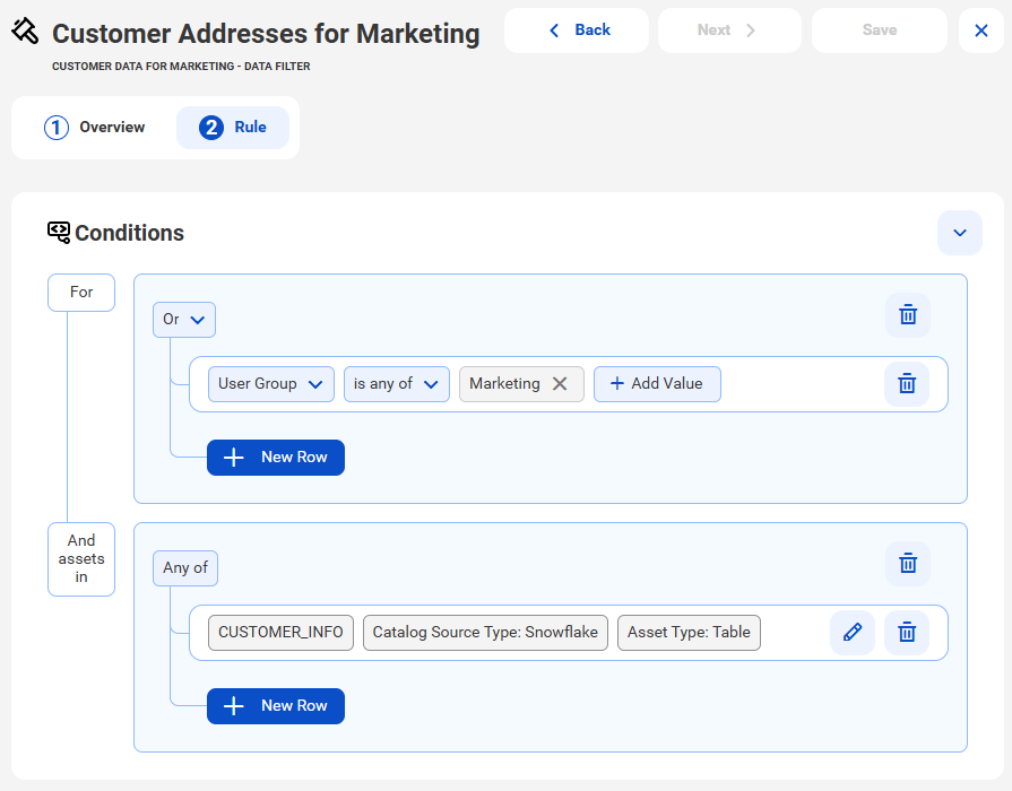


4. Click **Next**.

   The **Rule** tab appears. On this tab, specify the conditions under which Data Access Management will deny access to data records. The options in the **Conditions** section depend on whether the data filter policy that includes this rule uses the Data Integration/Data Marketplace enforcement method or the pushdown enforcement method.
   The following image shows the **Conditions** section of the **Rule** tab for the Data Integration/Data Marketplace enforcement method:



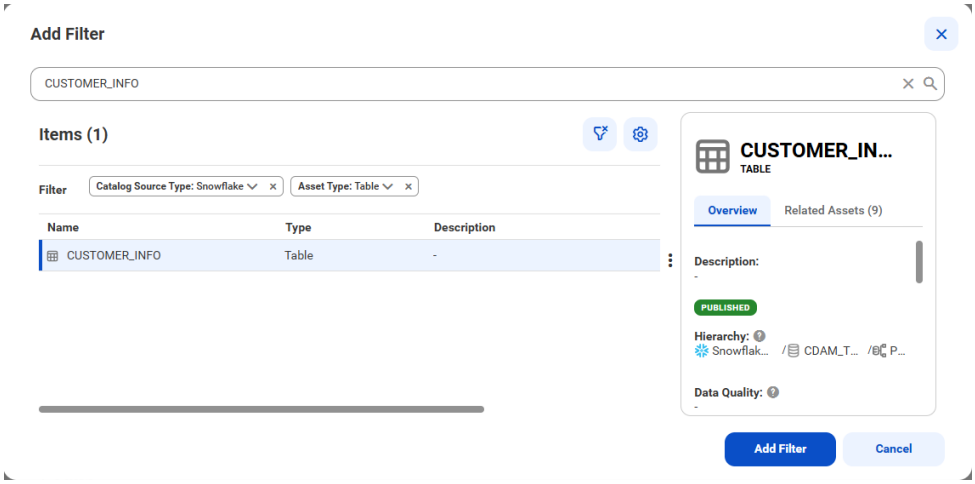
The following image shows the **Conditions** section of the **Rule** tab for the pushdown enforcement method:



5. In the **Conditions** section, click **New Row** in the **For** section.
6. For each condition, select an attribute, an operator, and relevant values.
   a. Select a contextual attribute, such as **User Group**.
   b. Select an operator, such as **is any of**.
   c. Click **Add Value** to select the value of the contextual attribute.
7. Click **New Row** again to create another condition.

8. For the pushdown enforcement type, click **New Row** in the **And assets in** section.
   a. Click **Add Filter** to select the catalog source type and data assets.

      The **Add Filter** window appears.
      The following image shows the **Add Filter** window:



   b. Enter a search query to find the data assets that you want to grant permissions to.

      For more information search queries, see *Asset Discovery* in the Data Governance and Catalog help.

      The **Preview** section shows the current results of that query. Adjust your search query to refine the results as needed.

      Note the following about query results:

      - The preview returns results based on currently accessible metadata. This might change with the addition of other assets that match the filter query.
      - Don't use the universal wildcard * because it grants access to all tables and views within the identified Data Governance and Catalog catalog source type.
      - A single permission assignment has a limit of 2,000 assets. If your query results in more than 2,000 assets, partition your filter across multiple rules to reduce the number of assets in the query.
      - For each data filter asset with a pushdown enforcement type, Data Access Management only processes the first 10,000 columns.

   c. Click the Filter icon.
   d. Click the **Add Filter** link, and select **Catalog Source Type**.

      Note: Data access control policies only work with specific catalog source types.

      For more information on catalog source types, see *Catalog Source Quick Reference* in the Data Governance and Catalog help.

   e. Select a catalog source type.
   f. Click **Add Filter** again, and select **Asset Type**.
   g. Select one or more asset types.
   h. Click the **Add Filter** button to return to the **Rule** section of the **Rule** tab.
   i. Click **New Row** again to create additional filters.

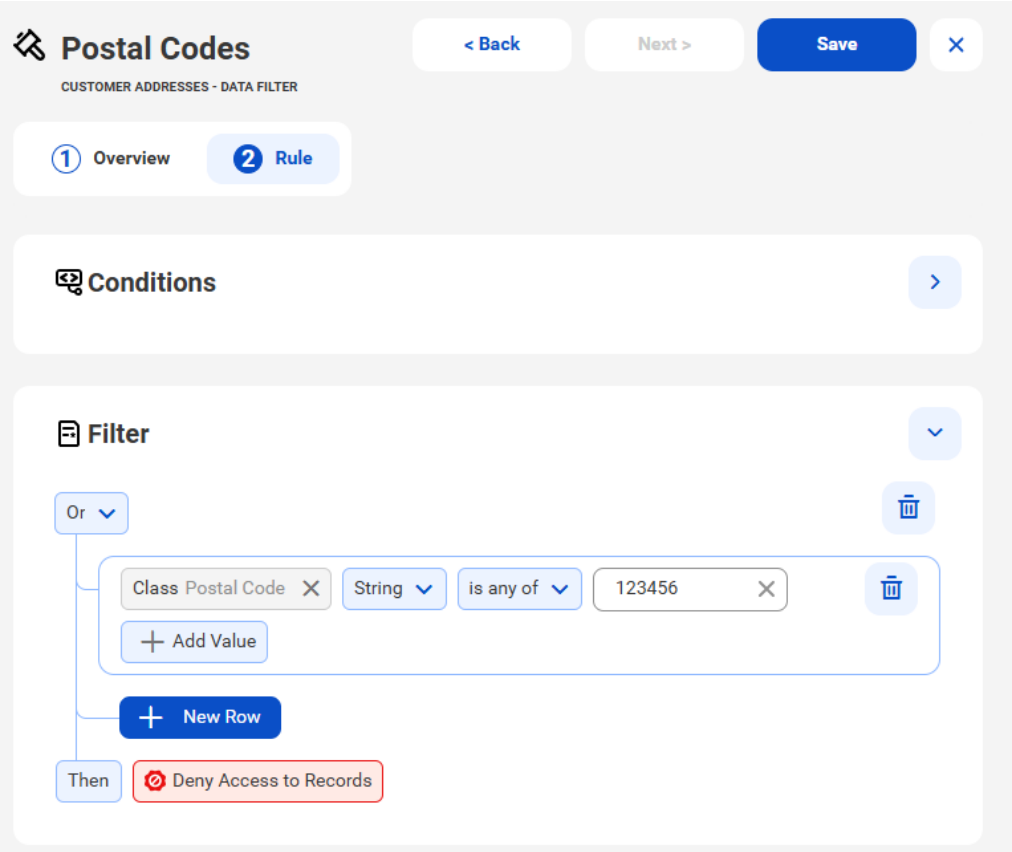
      You have now created conditions and filters for this rule.

Creating filters for data filter rules

After you create conditions for data filter rules, you create filters to deny access to data. Filters can determine if a rule activates when a single condition is true or when all conditions are true.

1. In the **Filter** section of the **Rule** tab, click **New Row**.

   The following image shows the **Filter** section of the **Rule** tab:

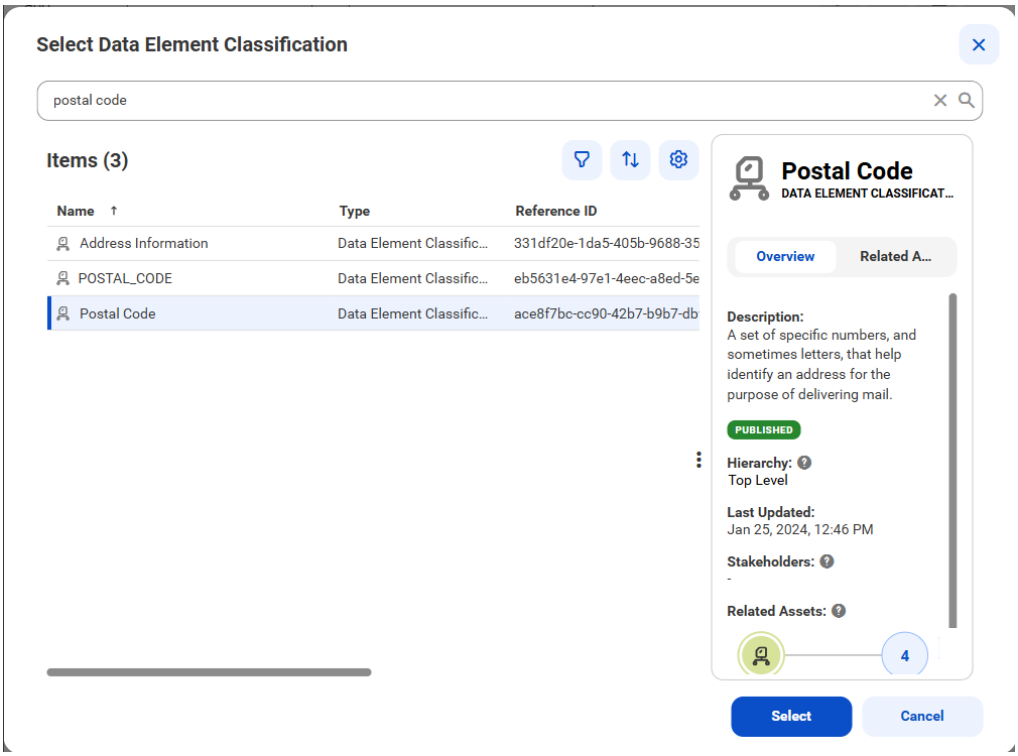


2. Click **Select Data Element Classification**.

   The **Select Data Element Classification** window appears.

   Note: If a data element has multiple data classifications, Data Access Management denies access to the row if any of the classifications matches the filter criteria.

The following image shows the **Select Data Element Classification** window:



3. Enter a search query to find the data assets.

   The **Preview** section shows the current results of the query. Adjust your search query to refine the results as needed.

4. Click a result and click **Select** to assign the data classification and return to the **Rule** tab.

5. Select a data type.

   Note: If you use data filter rules involving dates that might be applied in Access Policy transformations in Data Integration, create a data filter rule with two distinct criteria. In one criterion, use the date data type. In the other criterion, use the timestamp data type with the same values as the first criterion. This second criterion is for the Access Policy transformation.

6. Select an operator.

   Note: The data type you select determines list of available operators. Not all operators work with all data types.

7. Click **Add Value**.

   Note: Text strings are case-sensitive. The timestamp values that you enter are converted to and stored in Coordinated Universal Time (UTC).

8. Click **Add Value** again to add more values.

9. Click **New Row** to select another data element classification.

   a. Add the data element classification, data type, operator, and values as needed.

10. Continue this process until you have specified filters for the required data element classifications.

    Data Access Management denies access to a field if it meets any one of the filters.

11. Save the rule.

    Your updates take effect when the policy associated with the rule is published.
    If you do not have a workflow configured, the rule will automatically change to published status.

    If you have a workflow configured, the rule and its associated policy will change to draft status.

    For more information about designing workflows, see the *Administration* help in Metadata Command Center.

    Note: You can monitor the process to push down the policy initiated by using the data access sync job type on the **Monitor** page in Metadata Command Center.

    For more information about data access sync jobs, see the *Administration* help in Metadata Command Center.

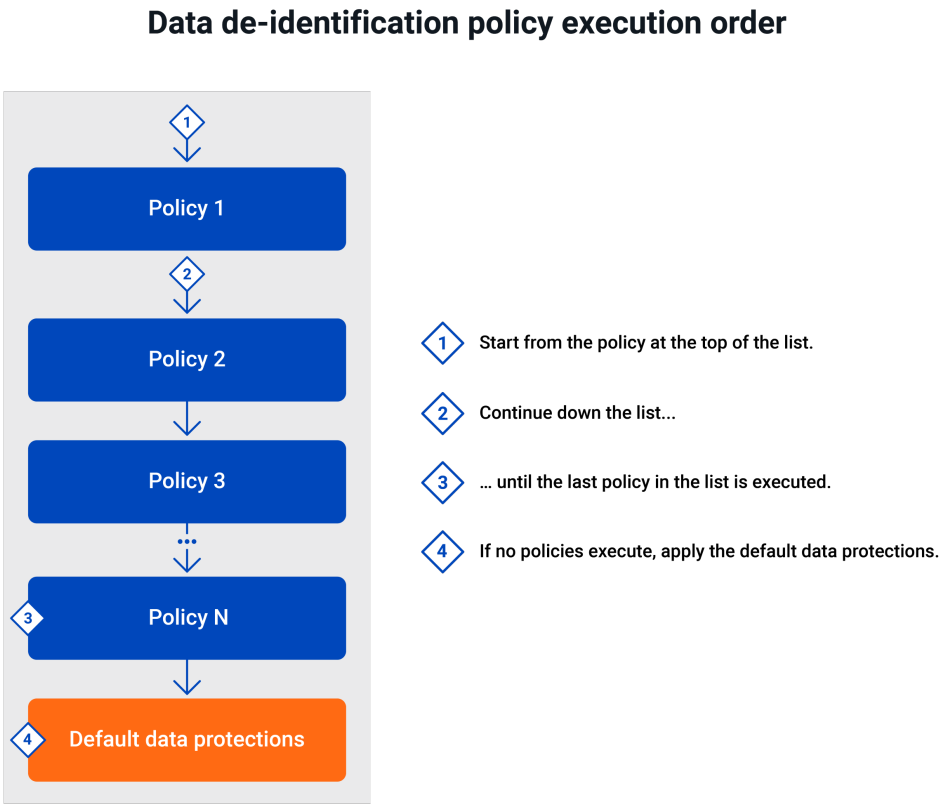
Default data de-identifications

Data Governance and Catalogapplies a default data de-identification technique if there are no data de-identification policies or the existing policies don't apply in a particular context or for particular classes of data.

If the field has a data element classification, Data Governance and Catalog applies the "Redact with null" on page 55 data de-identification technique.If the field does not have a data element classification, Data Governance and Catalog applies the "Retain" on page 55 data de-identification technique.

The default data de-identification technique also applies when Data Governance and Catalog can't apply a data protection to a specific field.

The following image shows the data de-identification policy execution order:



Data de-identification policy behavior

Data Governance and Catalog processes data de-identification policies from top to bottom.

Data de-identification policies handle data classifications in the following ways:

- If a data element has multiple data classifications, the first occurrence of any of the assigned data classifications determines the data protection applied.
- If a cell-level condition contains any of the multiple data classifications for the data element, then the cell-level condition is true and applies.
- If there is a cell-level condition using both data classifications for a data element with an **Or** clause, that condition evaluates to true, and the associated actions are applied.
- If there is a cell-level condition using both data classifications with an **And** clause, that condition evaluates to false when the combined conditions are *not met*.
- If there is a cell-level condition using both data classifications with an **And** clause, that condition evaluates to true when the combined conditions are met.
- Cell-level de-identifications take precedence before field-level de-identifications.

To learn more about how Data Governance and Catalog processes data de-identification policies and rules, see "Guidelines for sequencing data access policies for application" on page 15.

Unmasking behavior

You can unmask data that a user previously de-identified in IDMC.

When you prepare to unmask de-identified data, consider the following conditions:

- You can unmask data that a user de-identified using consistent tokenization and that uses the same consistency seed.
- If Data Governance and Catalog can't unmask the data, it returns the previously masked value.
- Unmasking works on field-level de-identifications.

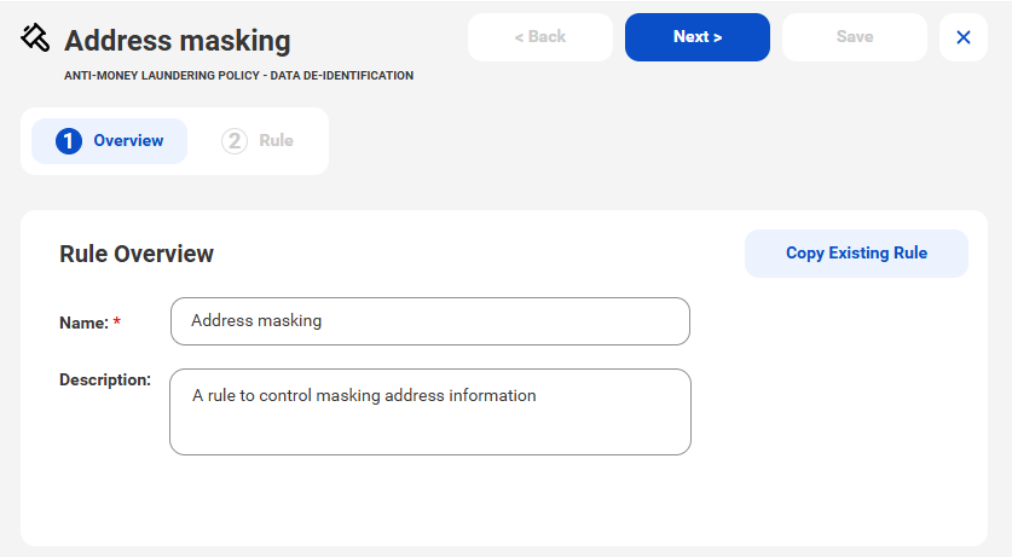
For more information about unmasking data, see the *Transformations* help in Data Integration.

Creating data de-identification rules

To define data de-identification rules, you add conditions based on user, usage, or metadata context, then assign data protections to data classes. Note that you create data protections separately on the Data Protection tab to complete all steps in this procedure.

1. Open a data de-identification policy and select the **Rules** tab.
2. Click the plus sign.

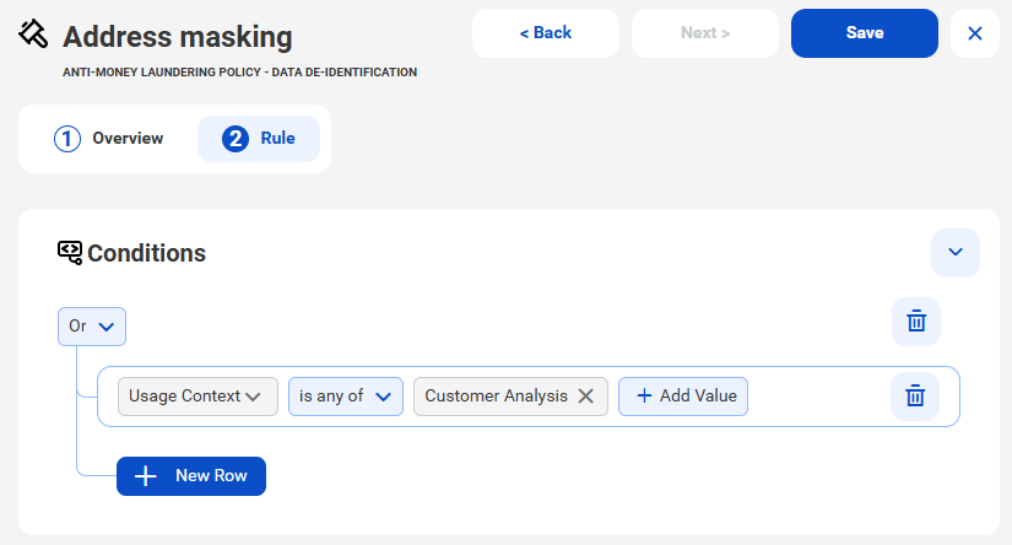
   The **Overview** tab appears.
   The following image shows the **Overview** tab:



3. Enter a name and description for the rule.
4. Click **Next**.

   The **Rule** tab appears. On this tab, you specify the conditions that will prompt Data Access Management to apply your data protections to data classes. You also specify the data protections.
5. In the **Conditions** section, click **New Row**.

The following image shows the **Conditions** section of the **Rule** tab:



6. Select **Or** or **And** to determine whether the rule activates when at least one condition is true ("Or") or when all conditions are true ("And").
7. For each condition, select an attribute, an operator, and relevant values.

   Each attribute appears.
8. Select a contextual attribute, such as **User Group** or **Usage Context**.
9. Select an operator, such as **is any of** or **is not any of**.
10. Click **Add Value** to select the value of the contextual attribute.
11. After you select the desired values within the list, click **Add Value** in the list to save those selections.
12. Click **New Row** again to create another condition.
13. Continue this process until you have created all of the conditions necessary.

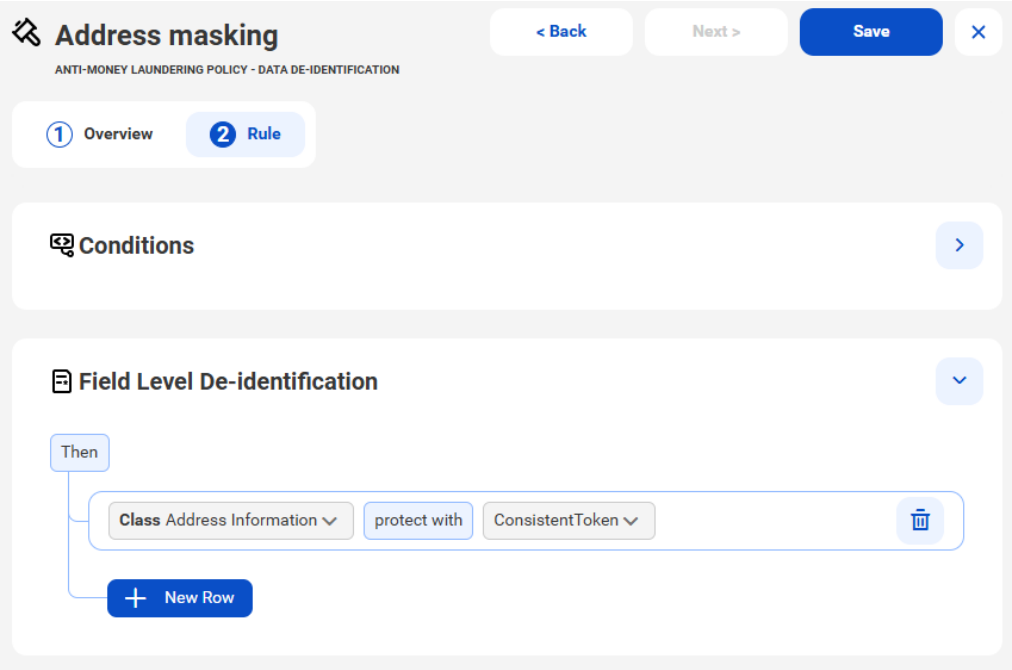
    You now create field-level de-identifications, cell-level de-identifications, or both for the rule. At least one field-level de-identification or cell-level de-identification is required.

Creating field-level de-identifications

After you create a data de-identification rule and add conditions to it, you can define field-level de-identifications. To complete all steps in this procedure, you'll create data protections separately on the Data Protection tab.

1. In the **Assign a data protection** section, click **Add Field-Level De-Identification**.

   The following image shows the **Field Level De-identification** section of the **Rule** tab:



2. Select a data class.
3. Assign a data protection to the data class.

   Note: You created a data protection separately on the Data Protection tab prior to this step to assign it in this step.
4. Click **New Row** to select another data class.

   Continue this process until you have specified all field-level data protections relevant to the rule.

   Note: If a data element has multiple data classifications, the first occurrence of any of the assigned data classifications determines the data protection applied.
5. Save the rule.

   Your updates take effect when the policy associated with the rule is published.
   If you do not have a workflow configured, the rule will automatically change to published status.

   If you have a workflow configured, the rule and its associated policy will change to draft status.

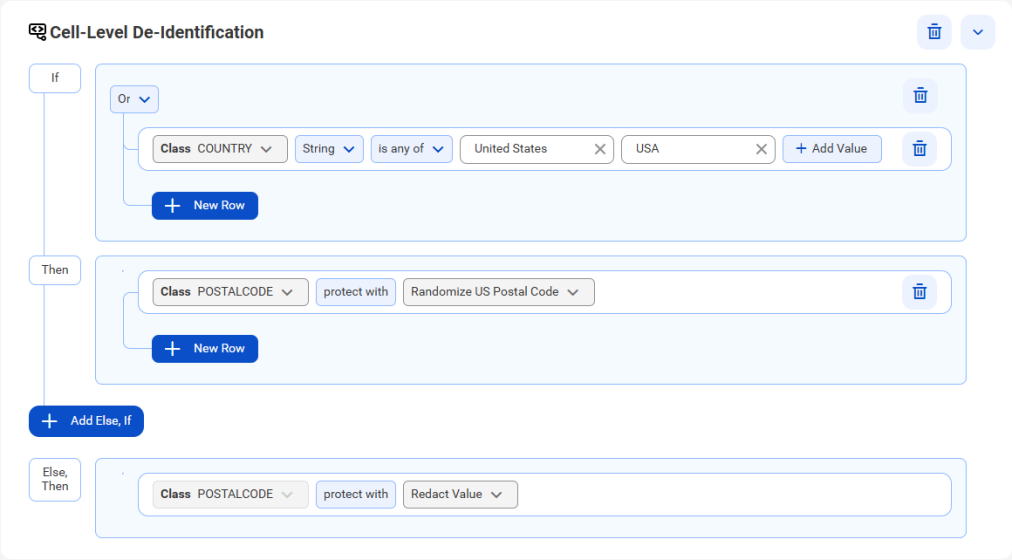
   For more information about designing workflows, see the *Administration* help in Metadata Command Center.

Creating cell-level de-identifications

After you create a data de-identification rule and add conditions to it, you can define cell-level de-identifications. To complete all steps in this procedure, you'll create data protections separately on the Data Protection tab.

1. In the **Assign a data protection** section, click **Add Cell-Level De-Identification**.

   The following image shows the **Cell-Level De-Identification** section of the Rule tab.



2. In the **If** section, select **Or** or **And** to determine whether the cell-level de-identification activates when at least one condition is true ("Or") or when all conditions are true ("And").
3. In the **If** section, click **New Row**.
4. Start creating a condition by selecting a data class.

   Note: When a condition contains multiple data classifications for the data element, if any of the data classifications meets the condition, the cell-level condition is true and applies.

   To learn more about how data de-identification policies process conditions, see "Data de-identification policy behavior" on page 40.
5. Select a data type.

   Note: If you use cell-level de-identification involving dates that might be applied in Access Policy transformations in Data Integration, create a cell-level de-identification with two distinct criteria. In one criterion, use the date data type. In the other criterion, use the timestamp data type with the same values as the first criterion. This second criterion is for the Access Policy transformation. You might need to adjust your filter or create multiple filters if your condition contains multiple constraints using an **And** operator.
6. Select an operator.

   Note: The data type you select determines list of available operators. Not all operators work with all data types.
7. Select or enter a value for the data class. Not all operators require values.

   Note: Text strings are case-sensitive. Your database will calculate the date value in Coordinated Universal Time (UTC).
8. Optionally, click **New Row** to add another condition to the data protection and repeat the previous steps.
9. In the **Then** section, select a data class to which to assign a data protection when the condition is true.

10. Select a data protection.

    Note: If a data element has multiple data classifications, the first occurrence of any of the assigned data classifications determines the data protection applied.

11. Optionally, click **New Row** to add another data protection to the condition.
12. You can optionally add an alternative condition and set of protections to your first condition by building a condition in the **Else, If** section:
    1. Select **Or** or **And** to determine whether the rule activates when at least one condition is true ("Or") or when all conditions are true ("And").
    2. Select a data class.
    3. Select an operator.
    4. Select or enter a value.
    5. Optionally, add a new row to add further data classes, operators, and values to the condition.
    6. In the **Then** section, assign a data protection to each data class.

       Note: The data classes that appear are the ones that you chose in the **If** statement. You can create multiple cell-level de-identification **Else, If** sections within a rule. Follow the same procedure for each **Else, If** section.

13. In the **Else, Then** section, specify how Data Access Management will transform the data classes that you identified as part of your initial **Then** condition that do not satisfy the conditions in this cell-level de-identification section.
14. Save the rule.

    Your updates take effect when the policy associated with the rule is published.
    If you do not have a workflow configured, the rule will automatically change to published status.

    If you have a workflow configured, the rule and its associated policy will change to draft status.

    For more information about designing workflows, see the *Administration* help in Metadata Command Center.

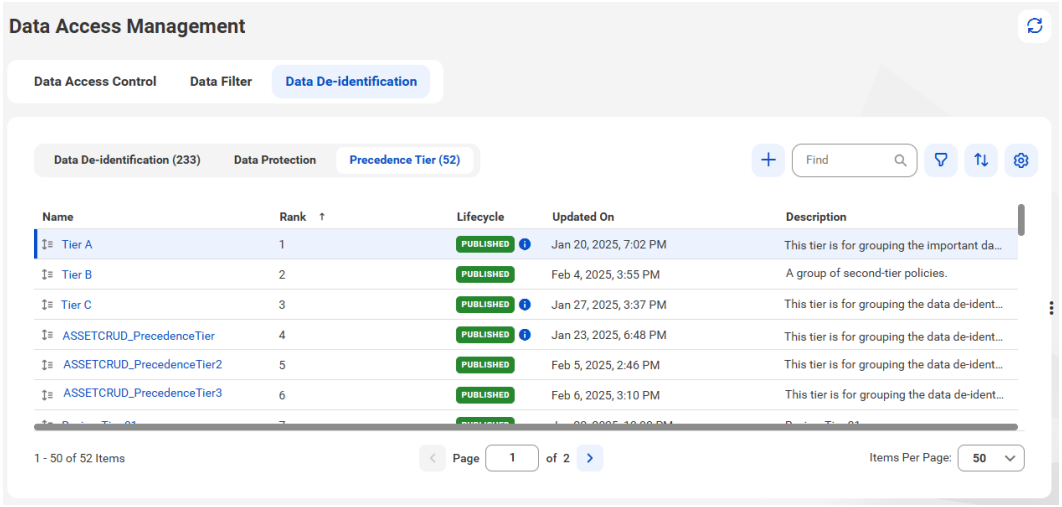
CHAPTER 3

Precedence tiers

Precedence tiers are ordered groupings of data de-identification policies that determine the priority and sequence of evaluating data de-identification policies and applying data protections. You assign data de-identification policies to a precedence tier.

Data Access Management evaluates the data de-identification policies within each tier from top to bottom. Data Access Management also evaluates precedence tiers based on rank from lowest integer to highest.

The following image shows the Precedence Tier tab:



Keep the following in mind when working with precedence tiers:

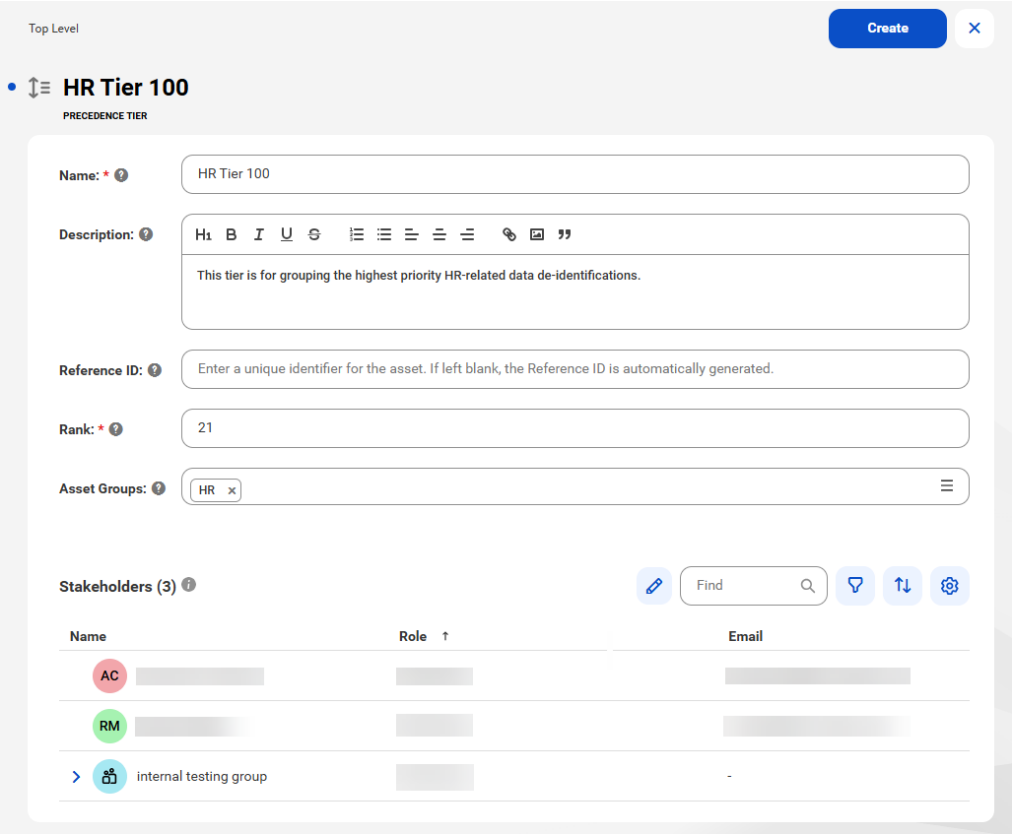
- You can't reorder data de-identification policies within a precedence tier if you don't have access to see all the policies within that precedence tier due to metadata access controls.
- You can't rank tiers if you can’t see all the precedence tiers.

Creating precedence tiers

You create precedence tiers to organize and prioritize data de-identification policies.

1. Click **Data Access**.
2. Click **Data De-identification**.
3. Click **Precedence Tier**.
4. Click the plus sign to open the New Precedence Tier page.

The following image shows the New Precedence Tier page:



5. Enter a title and description for the precedence tier.
6. Enter your own reference ID or leave the field blank automatically assign one.
7. Enter an integer as a rank for the precedence tier.

   The lower the number, the higher the rank.
8. Select Asset Groups to assign to this precedence tier.
9. Modify stakeholders for this precedence tier.
10. Click **Create**.

    The overview page for the precedence tier appears.

CHAPTER 4

Data protections

You can define data protections within a data de-identification policy's rules to protect data while maintaining data consistency, reversibility, and format.

Data de-identification techniques are behaviors for Data Access Management to apply to a field in a data set to de-identify it, while still preserving data utility.

You can use the following data de-identification techniques to define data protections:

- "Constant value" on page 51
- "Generalize date" on page 52
- "Hashing" on page 54
- "Redact with null" on page 55
- "Retain" on page 55
- "Substitute value" on page 56
- "Tokenize" on page 58
- "Truncate" on page 60

Creating data protections

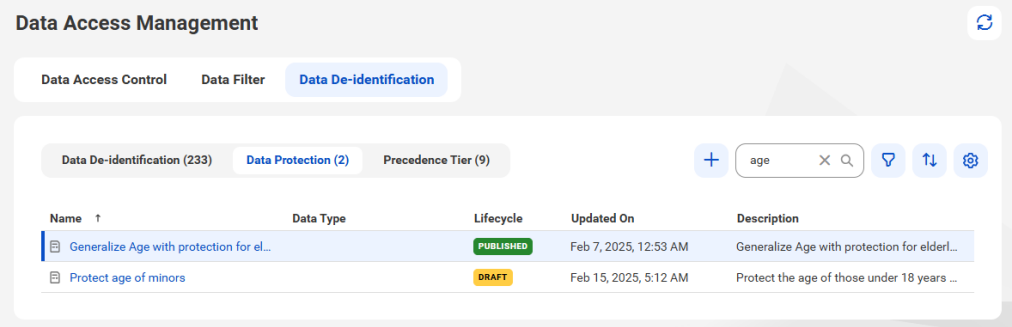
You create data protections and add them to data de-identification rules, which determine where and when to apply the data protections to data. You start by selecting a data type, then a relevant data de-identification technique, and finally defining how it transforms data.

A data protection can apply a different data de-identification technique per data type. For example, you can stipulate that the same data protection apply the truncate technique to a string, the tokenize technique to an integer, and the redact with null technique to unknown data types.

1. Click **Data Access** in the left navigation.
2. Click **Data De-identification**.
3. Click **Data Protection**.

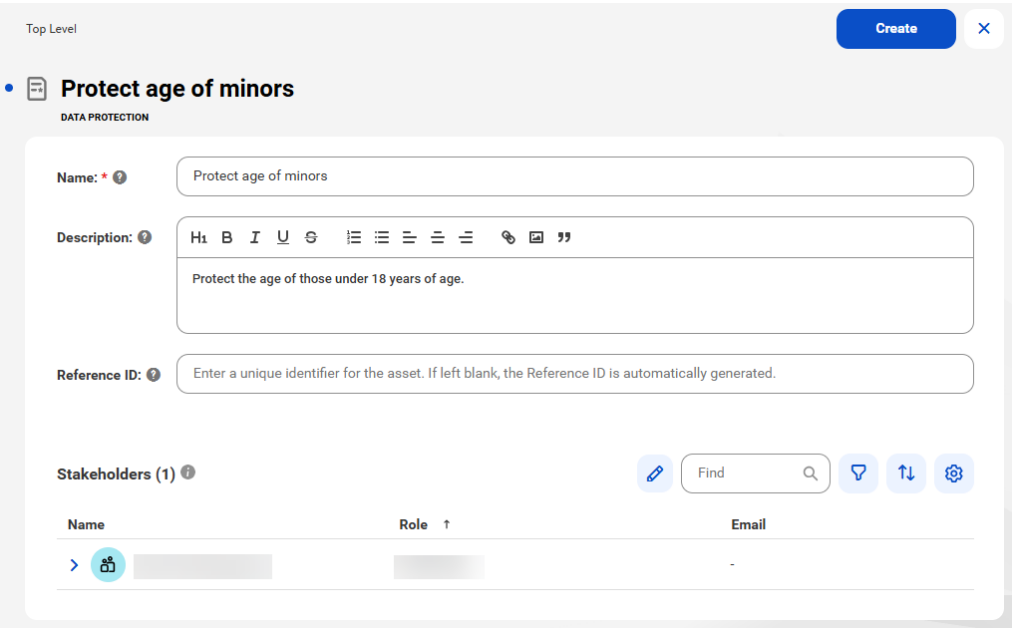
   A list of current data protections appears.

The following image shows the Data Protection tab:



4. Click the plus sign.

   The new data protection page appears.
   The following image shows the new data protection page:



5. Enter a name for the data protection.
6. Optionally enter a description, a reference ID, and select stakeholders.
7. Click **Create**.

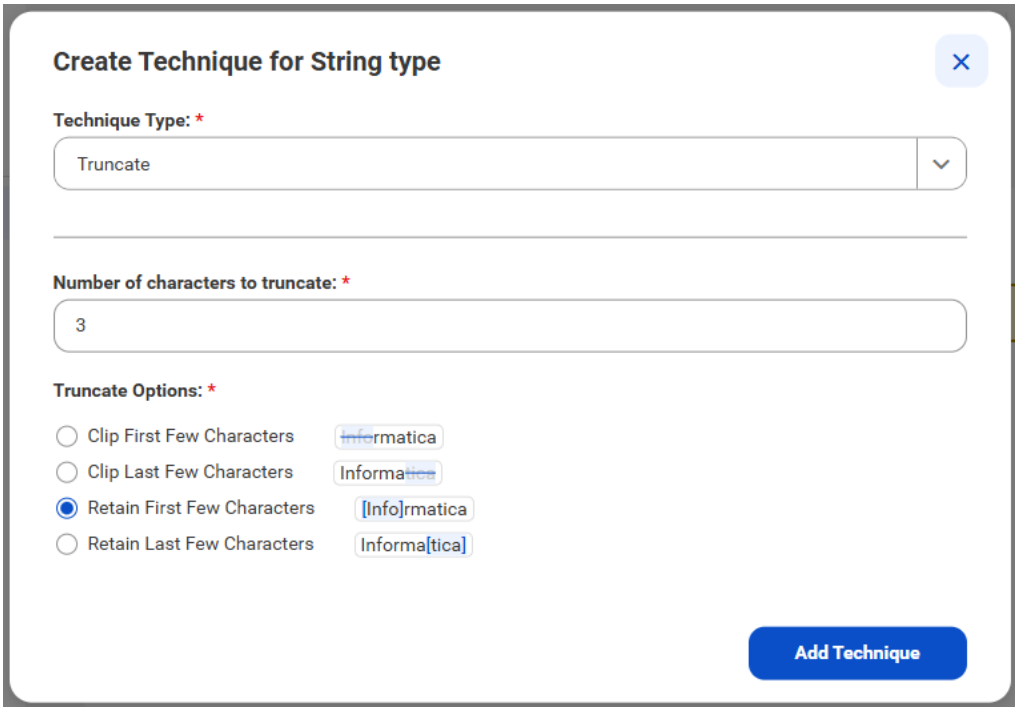
   The overview page appears.
8. Click **Techniques**.

   The Unknown/Any technique appears.
9. Click the plus sign to add data types relevant to this data protection.

   When you select a type, the **Create Technique** window appears.
10. Select a technique type.

    Fields appear to allow you to define the behavior of the data protection. The fields vary depending on the type of data de-identification technique that you select. See "De-identification techniques" on page 51 for an explanation of each data de-identification technique .

The following image shows an example of the fields used to define the behavior of one type of data protection:



11. Complete the relevant fields to determine the behavior.
12. Click **Add Technique**.
13. Click **Publish**.

    The data protection is now available to add to data de-identification rules.

Creating default data protections

Before you create any data de-identification policies, create default data protections for retaining values and redacting values to NULL.

1. Follow the instructions in "Creating data protections" on page 48.
2. Name one data protection "Retain Value."
3. Set this data protection for the "Any/Unknown" data type.
4. Assign the "Retain" on page 55 data de-identification technique to the data protection.
5. Create a second data protection with the name "Redact With Null."
6. Set this data protection for the "Any/Unknown" data type.
7. Assign the "Redact with null" on page 55 data de-identification technique to the data protection.

   You can now select these data protections when creating data de-identification rules.

De-identification techniques

Data de-identification techniques are behaviors for Data Governance and Catalog to apply to a field in a data set to de-identify it, while still preserving data utility. Learn about the available data de-identification techniques and see examples of each.

Constant value

Use the constant value data de-identification technique to mask all values of a given data class with the same value. The technique makes a permanent change in the target source system or query output and you can't unmask it.

The technique works with the decimal, integer, and string data types.

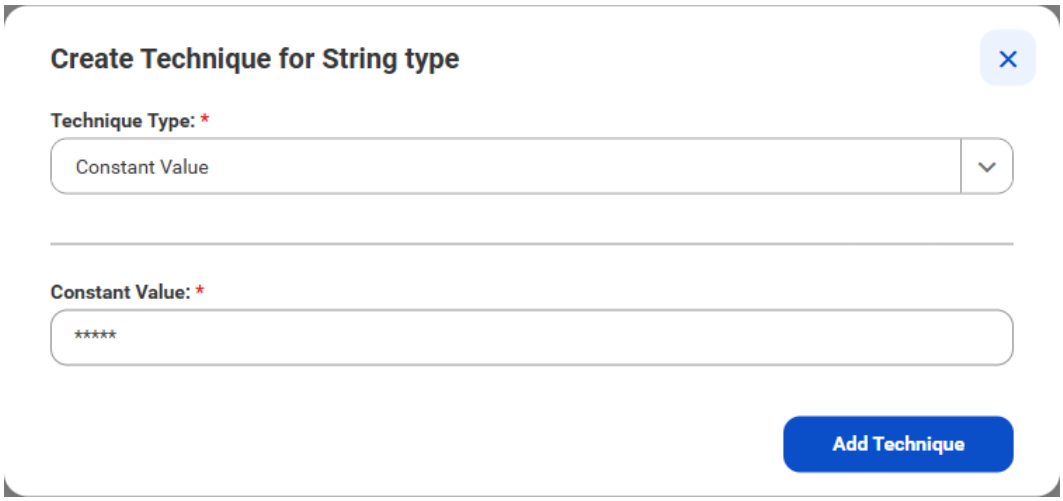
Note: The following behaviors apply to the constant value data de-identification technique:

- For decimal and integer data types, you can enter negative values.
- For decimal and integer data types, if you enter a constant value that is larger than the data source allows for the data type, the output is the upper limit of the allowable value. For example, PostgreSQL limits integers to 2,147,483,647. If you enter a constant value of 999999999999, the output will be 2147483647.

The following table lists examples of constant values that you might use to mask data:

Field	Data Type	Formats	Constant Value	Output
Password	string	- P@ssw0rd1 - MyP@$$w0rd!2345	********	********
Postal code	string	- 12345 - AB1 2CD	00000	00000
Salary	decimal	- 128354.89 - 75000.00	100500.50	100500.50
Geocoordinates	decimal	- 42.35843 - -71.05977	- 40.0 - -99.0	- 40.0 - -99.0
Age	integer	- 65 - 43	50	50

The following image shows the technique in the **Create Technique for String type** dialog box:



Generalize date

The generalize date data de-identification technique replaces the input date with a generalized version of the date or a constant date if the input date is outside of your defined date range. The technique makes a permanent change in the target source system or query output and you can't unmask it.

For timestamp and datetime data types, the format is HH:mm:ss, and the output time is always set to midnight (00:00:00). For example, 16-09-2018T11:33:00 becomes 01-01-2018T00:00:00 if the date is generalized to only preserve the year.

The technique works with the date and timestamp data types.

The following table describes the date generalization options that you can use in the **Generalize** section:

Option	Description
Preserve **Year** (YYYY)	Retains the year from the input values and generalizes the day and month to 01-01. Note that Data Governance and Catalog might still change the year depending on the settings that you select in **Protect elderly individuals** and **Protect minor or underage individuals**.
Preserve **Month and Year** (MM-YYYY)	Retains the month and year from the input values and generalizes the day to 01. Note that Data Governance and Catalog might still change the month and year depending on the settings that you select in **Protect elderly individuals** and **Protect minor or underage individuals**.
Preserve **Day, Month and Year** (DD-MM-YYYY)	Retains the day, month, and year from the input values. **Note:** When you select this option, also select **Protect elderly individuals** or **Protect minor or underage individuals**. Note that Data Governance and Catalog might still change the day, month, and year depending on the settings that you select in **Protect elderly individuals** and **Protect minor or underage individuals**.

The following table describes the date generalization options that you can use in the **Generalize user-defined date range** section:

Option	Description
Protect elderly individuals	Returns a constant date value when an input date is above an age threshold that you specify. You enter the threshold in years in the Age field. You can use the option to obscure the birth dates of individuals who are older than an age that you enter. If the difference between the input date and the current date is greater than or equal to the age threshold, Data Governance and Catalog returns a constant date. Data Governance and Catalog generates the constant date dynamically as the date of data provision minus the number of years in the Age field. If the difference between the input date and the current date is less than the age threshold, Data Governance and Catalog returns the input date with no change. **Note:** The Generalize Date option works in conjunction with any Preserve option that you set. If the date difference is greater than or equal to the age threshold, the Generalize Date option takes precedence over the Preserve option. If the date difference is less than the age threshold, Data Governance and Catalog applies the Preserve option and does not generate a constant date.
Protect minor or underage individuals	Returns a constant date value when an input date is below an age threshold that you specify. You enter the threshold in years in the Age field. You can use the option to obscure the birth dates of individuals who are younger than an age that you enter. If the difference between the input date and the current date is less than or equal to the age threshold, Data Governance and Catalog returns a constant date. Data Governance and Catalog generates the constant date dynamically as the date of data provision minus the number of years in the Age field. If the difference between the input date and the current date is greater than the age threshold, Data Governance and Catalog returns the input date with no change. **Note:** The Generalize Date option works in conjunction with any Preserve option that you set. If the date difference is less than or equal to the age threshold, the Generalize Date option takes precedence over the Preserve option. If the date difference is greater than the age threshold, Data Governance and Catalog applies the Preserve option and does not generate a constant date.

The **Protect elderly individuals** and **Protect minor or underage individuals** options can help you comply with the U.S. Health Insurance Portability and Accountability Act (HIPAA) requirement for the storage of ages and dates contained in patient healthcare data.

For example, if you select "Preserve **Month and Year** (MM-YYYY)," and you specify to generalize the birth date for all individuals older than 90 years old, this would result in the following:

If Data Governance and Catalog transforms the data on December 31, 2022, then Data Governance and Catalog will generalize any date that is after December 31, 1932 (2022-90=1932) to 01-[original month value]-[original year value]. Data Governance and Catalog will generalize any date that is before December 31, 1932 to 01-12-1932. For example, Data Governance and Catalog would change the input date of June 2, 1929 (02-06-1929) to 01-12-1932.

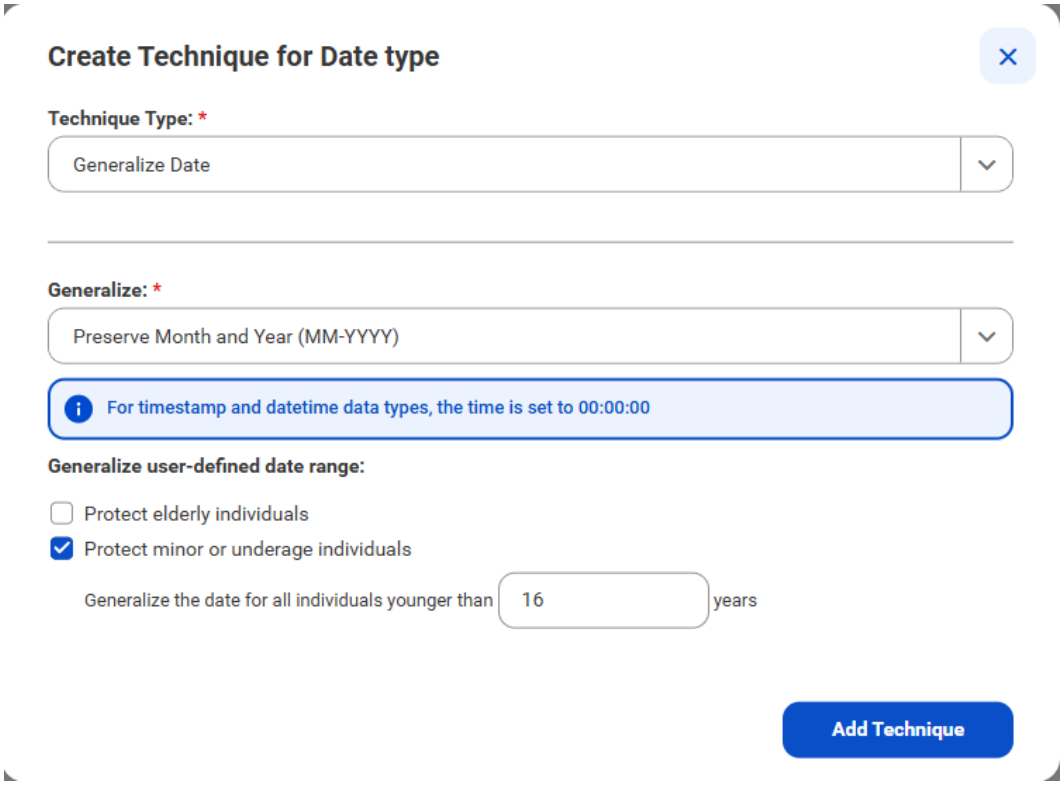
Data Governance and Catalog would not generalize any date that is equal to December 31, 1932.

Alternatively, if you select "Preserve **Month and Year** (MM-YYYY)," and you specify to generalize the birth date for all individuals younger than 16 years old, this would result in the following:

If Data Governance and Catalog transforms the data on December 31, 2022, then Data Governance and Catalog will generalize any date that is before December 31, 2006 (2022-16=2006) to 01-[original month value]-[original year value]. Data Governance and Catalog will generalize any date that is after December 31, 2006 to 01-12-2006. For example, Data Governance and Catalog would change the input date of June 2, 2007 (02-06-2007) to 01-12-2006.

Data Governance and Catalog would not generalize any date that is equal to December 31, 2006.

The following image shows the technique on the **Create Technique for Date type** window:



Hashing

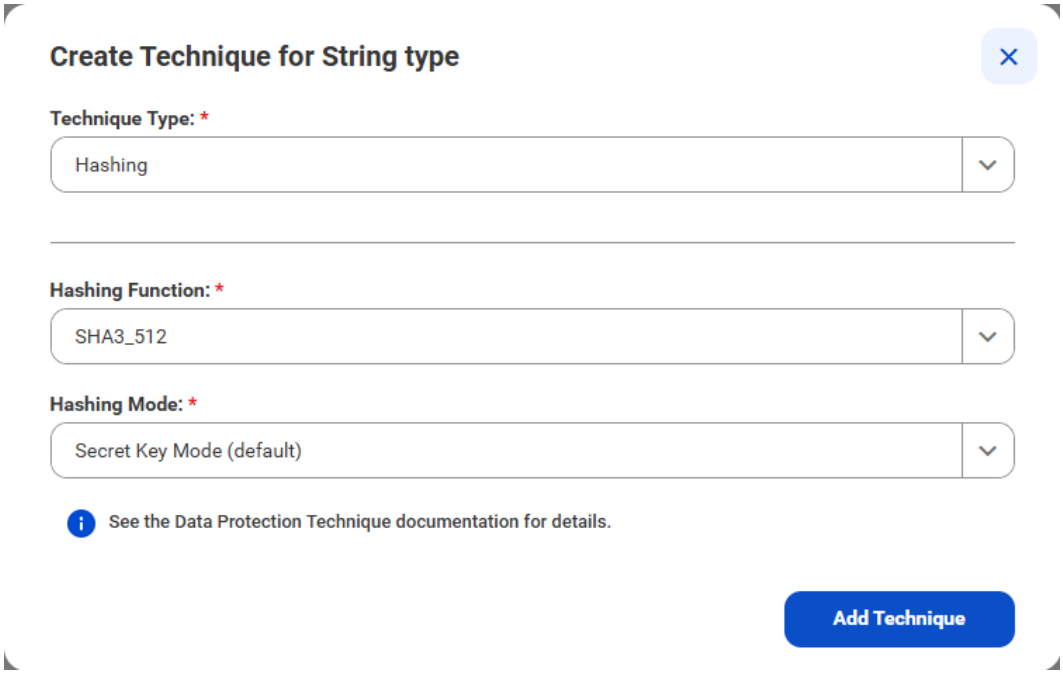
The hashing data de-identification technique replaces input values with hashed values. You can configure the data protection to use one of multiple hashing functions. The technique makes a permanent change in the target source system or query output. The data can't be unmasked.

The hashing data de-identification technique works with the string data type.

The following table describes the fields you use to configure a hashing data protection:

Option	Description
Hashing Function	Specifies the function used to hash the values.
Hashing Mode	Specifies which mode to use when hashing values. "Secret Key Mode (default)" combines the input value with a secret key before hashing the outputs. This is known as "adding pepper" to the values. Use this mode if your organization does not need to combine these values with previously hashed values. "Compatibility Mode" uses the hash function with no secret key to be compatible with other systems. Choose this mode if your organization needs to combine these values with previously hashed values.

The following image shows the technique on the **Create Technique for String type** window:



Redact with null

Use the redact with null data de-identification technique to replace all field values in a column with null. All users see the column. The technique makes a permanent change in the target data source or query output and you can't unmask it.

The technique works with all data types.

The technique is useful when you want to remove field values while preserving the schema of the asset after data transformation.

Retain

Use the retain data de-identification technique to retain field values. No masking behavior takes place.

The retain data de-identification technique works with all data types.

Substitute value

The substitute value data de-identification technique replaces input values with human-readable substitutes that you provide. In some use cases, you might prefer these substitutes to tokenized values. You can configure the data protection to substitute values consistently or randomly. The technique makes a permanent change in the target source system or query output. The technique makes a permanent change in the target source system or query output and you can't unmask it.

The substitute value data de-identification technique works with the string data type. Regardless of the content of a value, the technique treats it as a string. The technique works with Unicode characters.

When you create the substitute values file, organize it as follows:

- Create a single list of values in one column.
- Do not include a header row.
- Separate each value with a line break.

The following table lists the fields you use to configure a substitute value data protection:

Option	Description

File Name	Specifies the name of the file that includes the substitution values.
	To make the substitution values file available for use with Data Integration, verify that the file is in the following location where the Secure Agent is installed:
	`<Secure Agent installation directory>\apps \Data_Integration_Server\data`
	To make the substitution values file available for use with Data Marketplace, verify that the file is in the following location where the Secure Agent is installed:
	`<Secure Agent installation directory>\apps \Data_Access_Management_Proxy\conf\vaultless-lookup-dictionaries`

Consistency Behavior	Specifies whether to substitute values consistently or randomly. "Consistently Substitute" means that the same substitute value always replaces the same input value. However, the same substitute value might replace more than one input value. The cardinality of the input set and the size of the substitution list determines the likelihood that the same substitute value replaces more than one input value.

In consistent substitution, substitute values replace input values consistently but not uniquely.

The following table illustrates this concept with an example of using the "Consistently Substitute" option with possible substitution values of James, Jonathan, Jose, Karim, Briana, and Luanne:

Input Values	Output Values
Rodrigo	Briana
Rodrigo	Briana
Kim	Jose
Kim	Jose

Input Values	Output Values
Carl	Briana
Carl	Briana

Note that Briana always substitutes Rodrigo and Jose always substitutes Kim. However, Briana also always substitutes Carl. Therefore, Briana is not unique, but it consistently replaced input values.

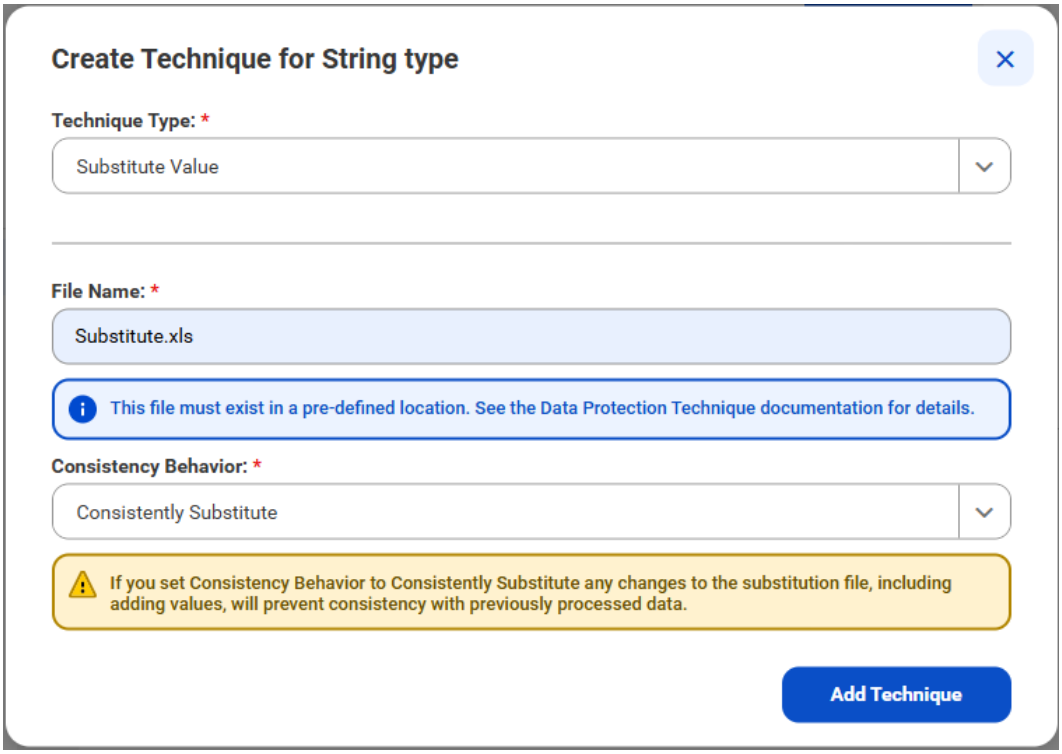
In random substitution, each row is distinct from others.

The following table illustrates an example of using the "Randomly substitute values" option with possible substitution values of James, Jonathan, Jose, Karim, Briana, and Luanne:

Input Values	Output Values
Rodrigo	Briana
Rodrigo	Karim
Kim	Briana
Kim	James
Carl	Luanne
Carl	Karim

Note that while Briana substitutes Rodrigo in the first row, any possible substitution value can replace Rodrigo in the second row.

The following image shows the technique on the **Create Technique for String type** window:



Note: For details on using the **File Name** field, see the "File Name" row in the table that lists the fields you use to configure a substitute value data protection at the beginning of this section.

Tokenize

The tokenize data de-identification technique replaces sensitive data with randomly generated equivalents, called tokens. Tokens are synthetic, alphanumeric values that stand in for the original sensitive data. You define the input value and output token format using the regular expression syntax.

Note: To create tokens, Data Governance and Catalog uses the FF1 format-preserving encryption cipher. This cipher is based on the Advanced Encryption Standard (AES) cipher.

The tokenize data de-identification technique works with the string and integer data types. The regular expression syntax specifies a pattern for the generated string.

For example, to replace the value abcdef with a randomly generated six-letter string, use the regular expression syntax [a-z]{6}. This produces the value mvskyc.

To replace the initial value 1234 with a randomly generated four-digit number, use the regular expression syntax [1-9]{4}. This produces the value 4159.

Define an expression whose minimum and maximum integer values are compatible with the data type. For example, if your input data type stores numbers from -100 to 100, define an expression that limits output to that range.

The following table describes the fields you use to configure a tokenize data protection:

Option	Description
Regular Expression	The syntax that specifies a pattern for the generated string. For details on the regular expression-style syntax, see Chapter 7, "Tokenization syntax reference" on page 67.
Consistency Behavior	**Consistently Tokenize** ensures that Data Governance and Catalog always replaces the same input value with the same tokenized value. **Randomly Tokenize** ensures that Data Governance and Catalog always replaces the same input value with unique tokenized values. Data Governance and Catalog doesn't preserve data consistency and might duplicate tokens.

With **Consistently Tokenize**, Data Governance and Catalog replaces raw text or numeric values with a tokenized value derived from the encrypted value of the input. In this case, the application of a tokenize data de-identification technique to tokenize data is *consistent*, meaning that the same input value always results in the same tokenized value whenever the same rule processes it.

With **Randomly Tokenize**, Data Governance and Catalog replaces raw text or numeric values with randomly generated tokenized values. In this case, the application of a tokenize data de-identification technique to tokenize data is *inconsistent*, meaning that token duplication is allowed. There will likely be multiple tokenized values for the same input value if it occurs more than once in the data.

Using **Consistently Tokenize** is required if a keyed relationship, such as maintaining referential integrity, is to be maintained across multiple de-identified tables. For rules that you configure to mask or tokenize consistently, you might optionally allow for unmasking of data that was de-identified through that rule.

Note: When using consistent tokenize, Data Governance and Catalog expects all data values to match the regular expression. If all the data values do not match the regular expression, the request will fail and return an error. For example, if a tokenize data de-identification technique specifies a capital letter, for example [A-Z][a-z]{1,15}, and the data doesn't include a capital letter, Data Governance and Catalog returns an error.

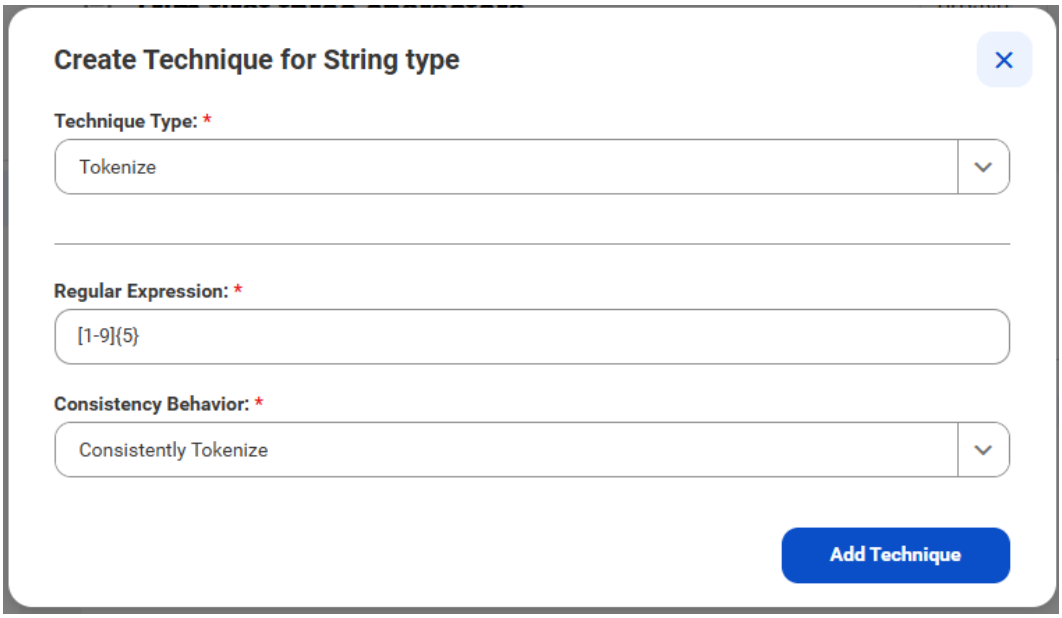
When creating tokenize data de-identification techniques, you can use any UTF-8 character block. For example, Data Governance and Catalog works with the regular expression [a-z]{10}[ก-ฅ]{10}, which includes both Basic Latin and Thai characters.

However, you can't mix UTF-8 blocks within the same regular expression component. For example, you can't use both Basic Latin and Thai characters in the same regular expression component, for instance [a-ก]{3}.

The following table lists examples of regular expressions that you can use to replace the value in a given format:

Field	Sample Input	Expression
Email Address	williamjohnson@samplecompany.com	[a-z]{7}\@[a-z]{5}\.com
Last Name	johnson	[a-z]{15}
Employee ID (For example, 52938486)	12345678	[1-9]{8}
U.S. ZIP Code (For example, 93612)	12345	[1-9]{5}

The following image shows the technique on the **Create Technique for String type** window:



Truncate

The truncate data de-identification technique removes part of the input value. You can configure the data protection to start from the beginning or end of the value. By default, the technique starts from the left. The technique makes a permanent change in the target source system or query output and you can't unmask it.

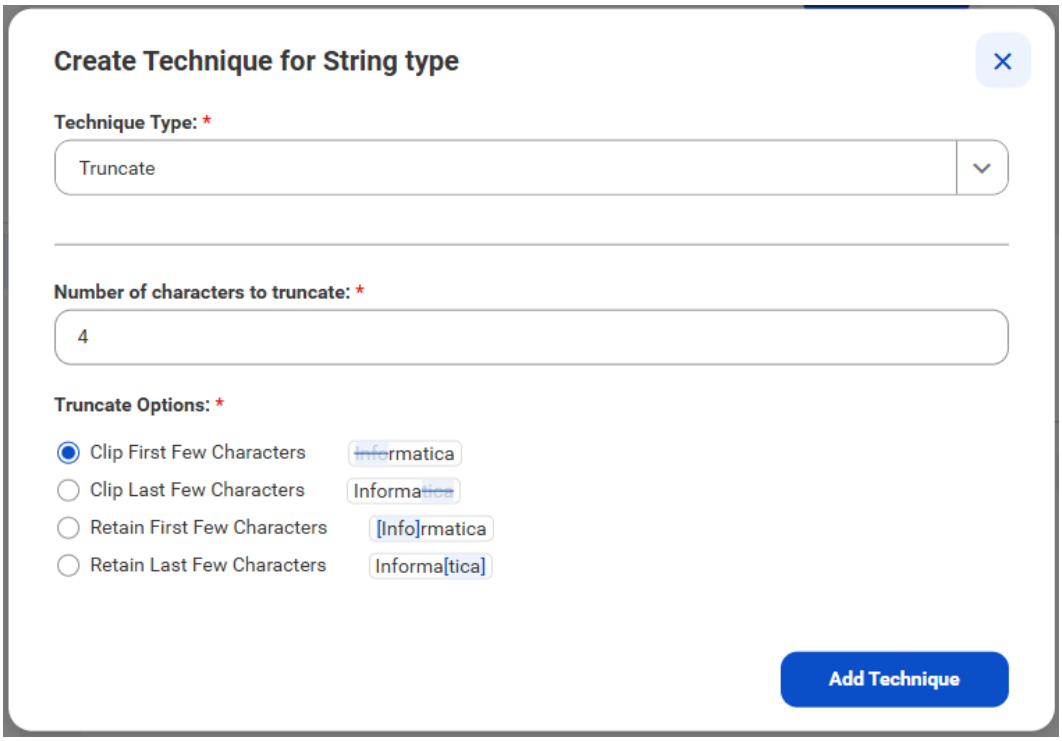
The truncate data de-identification technique works with the string data type.

The following table lists the fields you use to configure a truncate data protection:

Option	Description
Number of characters to truncate	Specifies the number of characters to remove or preserve in an input value.
Truncate Options	Select **Clip First Few Characters** to remove characters from the left side of the string. Select **Clip Last Few Characters** to remove characters from the right side of the string. Select **Retain First Few Characters** to retain characters from the left side of the string. Select **Retain Last Few Characters** to retain characters from the right side of the string.

For example, you want to remove three characters from the United Kingdom post code SE1 8DJ. If you specify three characters, starting from the left, and you select **Preserve characters**, Data Governance and Catalog returns SE1 for the post code. If you specify three characters, starting from the left, and you don't select **Preserve characters**, Data Governance and Catalog returns 8DJ for the post code.

The following image shows the technique on the **Create Technique for String type** window:



CHAPTER 5

Data access asset lifecycle

After you create a data access asset, such as a data access policy, a precedence tier, or a data protection, you can publish it. When you edit a published data access asset, Data Access Management creates a draft copy of it. The published version remains active until you publish the draft version.

Data access assets can have the following statuses:

- Draft: A user has not yet published the data access asset or there is a new draft copy of the data access asset following an edit.
  It is inactive.
- Published: A user has published the data access asset.
  It is active if other criteria are also true. Depending on the asset type, these criteria include effective dates, enabled, and policy conditions.

To manage the lifecycle of data access assets and ensure users review changes, enable workflow.

For more information about configuring a workflow event, see *Administration* in the Metadata Command Center help.

CHAPTER 6

Audit events

Audit events are detailed explanations of events such as the creation of a data protection and data access policy updates. You can view some audit events on the **History** tab of any data access asset on the Data Access Management page in Data Governance and Catalog. An administrator at your organization can set the Secure Agent to write other audit events to a local directory path. These events include query events and data access control policy pushdown events.

You can use audit events to track data processes and data use. For example, you can identify the users who created and modified data access assets.

Data access asset history

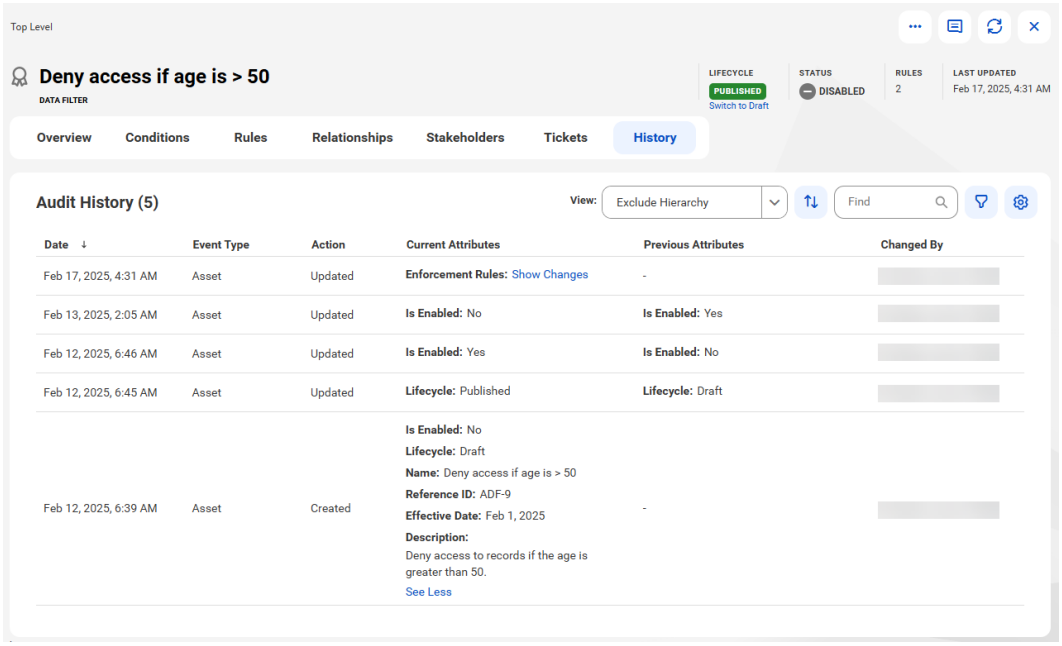
On the **History** tab of any data access asset on the Data Access Management page in Data Governance and Catalog, you can view changes that users have made to the asset.

The **History** tab provides an audit log that you can use for compliance requirements.

The following table describes the fields that appear on the **History** tab:

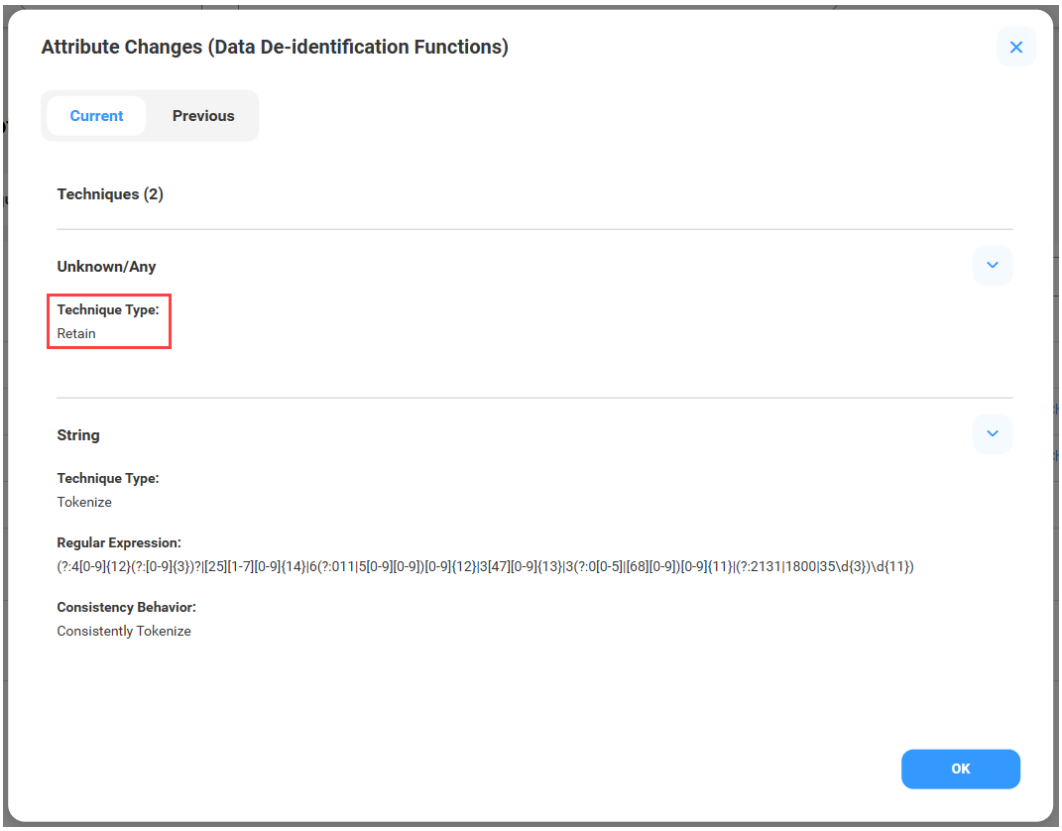
Field	Description
Date	Date on which the change was made.
Event Type	The event type is always "Asset" for data access assets.
Action	What action the user performed on the data access asset. Examples include "Created" and "Updated."
Current Attributes	Value in the field after the change. When available, click **Show Changes** for details.
Previous Attributes	Value in the field before the change. When available, click **Show Changes** for details.
Changed By	User that made the change.

The following image shows the **History** tab of a data access asset:

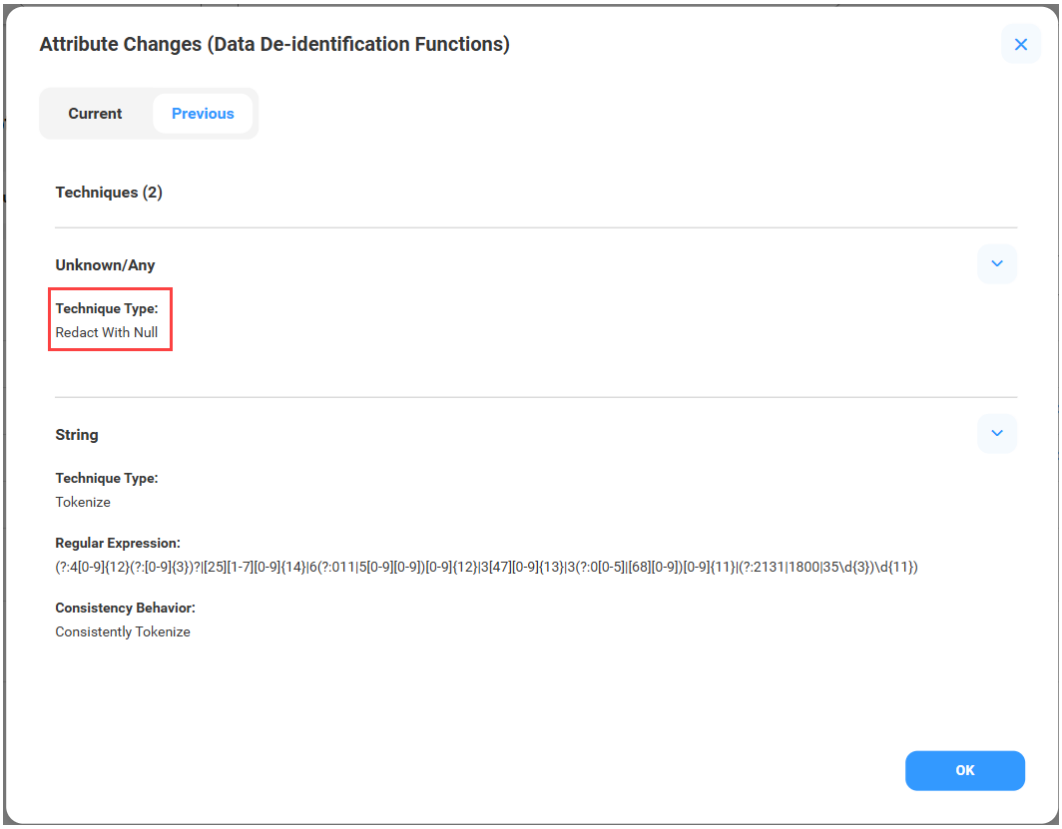


When available, click **Show Changes** in the **Current Attributes** or **Previous Attributes** fields, to view detailed changes.

The following image shows the **Current** tab of the **Attribute Changes** window:



The following image shows the **Previous** tab of the **Attribute Changes** window:

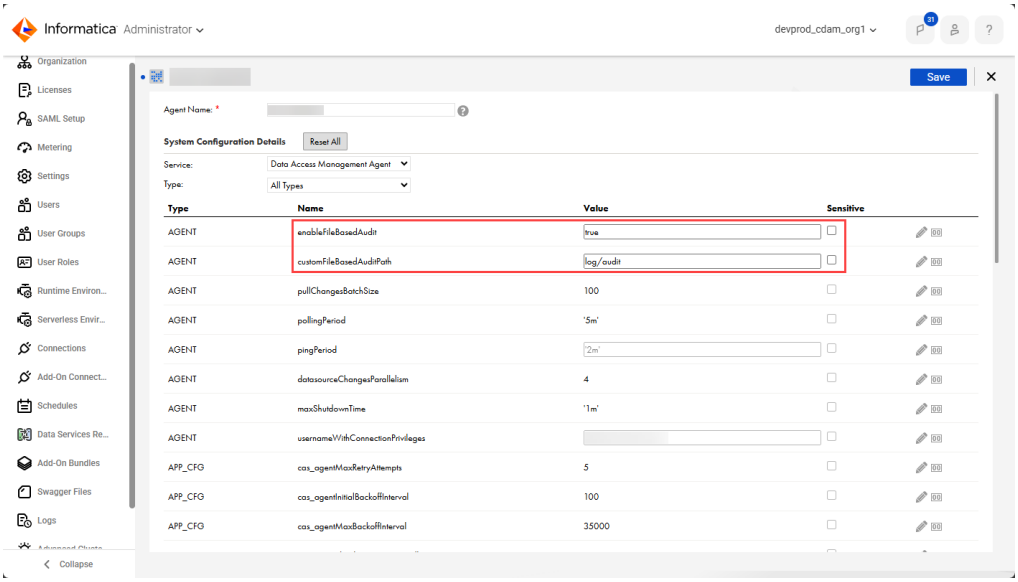


Writing audit events to disk

An administrator at your organization can set the Data Access Management Agent service and the Data Access Management Proxy service in the Secure Agent to write audit events to a local directory path. These events include query events and data access control policy pushdown events.

1. Access the Administrator service.
2. Click **Runtime Environments** in the left navigation.

3. Click the name of your Secure Agent.



4. Click **Edit**.
5. Scroll down to the **System Configuration Details** section.
6. Select Data Access Management Agent in the **Service** menu.
7. Set enableFileBasedAudit to "true."
8. In customFileBasedAuditPath, enter a directory path where the Secure Agent writes audit logs. The default path is the standard output stream (stdout).
9. View the properties of the Data Access Management Proxy in the **Service** menu.
10. Set enableFileBasedAudit to "true."
11. In customFileBasedAuditPath, enter a directory path where the Secure Agent writes audit logs. The default path is the standard output stream (stdout).
12. Click **Save**.

    To learn more about these services, see *Secure Agent Services* in the Administrator help.

    For more information about these services, see *Secure Agent Services* in the Administrator help.

CHAPTER 7

Tokenization syntax reference

You use a regular expression-style syntax to define the format of the input values and output tokens that Data Governance and Catalog processes when you use the tokenize data de-identification technique. Understand the limitations and follow examples of syntax that works with Data Governance and Catalog.

Note: Since you only use the tokenization expression to define and match the format of a string, and not for extracting portions of it, the tokenization syntax that you use does not support all of the constructs available in other regular expression syntaxes.

The following are examples of what you can't use in tokenization expressions:

- Anchors
- Backreferences
- Capture groups
- Repeated elements with unbounded repetitions
- These shorthand character classes: /d, /D, /h, /H, /s, /S, /v, /V, /w, /W, and .

Follow this guide when constructing your tokenization expression.

Unicode character support

The regular expression syntax that Data Governance and Catalog uses natively works with Unicode characters.

Note the following about using Unicode characters:

- Not all Unicode characters are printable.
- In character ranges, Data Governance and Catalog interprets the endpoints as points in the Unicode character table. Data Governance and Catalog forms a continuous range of all the characters between the endpoints. For example, the expression [A-z] includes the characters: [\]^_` as well as all upper- and lower-case letters.
- In character ranges, if you use an endpoint that comes before the start point in a Unicode table, the expression will produce no values.

Any character

You use the period or dot character to represent any Unicode character in an expression.

For example, the expression .{5} will produce tokens that include *any* five Unicode characters. Examples include:

- 些A 筈
- 8 塿'칲
- ^堇嫩

Character ranges

Use square brackets to define a character class. Data Governance and Catalog will choose one of the available options.

For example, the expression [abc] produces any of the following tokens:

- a
- b
- c

Within a character class, you can specify a character range with a dash. The range represents all of the characters between and including the start and end of the range. For example, both of the following expressions will produce a token of any letter from the English alphabet:

- [a-z]
- [abcdefghijklmnopqrstuvwxyz]

You can include multiple character ranges in a single character class. For example, the expression [A-Za-z0-9] will produce a token comprised of a single character that is either any English upper- or lower-case letter or any digit.

You can use the caret symbol to negate character classes. For example, the expression [^X] produces all possible Unicode characters, except for X. This is useful when you use it in combination with the intersection operator.

Complement operators

You can create expressions with intersections that exclude select values. To do this, you use a tilde character (~) to introduce a complement operator to an expression.

For example, the expression ([0-9]{3})&~(111) can produce any three-digit number between 000 and 999, except for 111.

The & indicates an intersection.

For more information about intersections, see “Intersections” on page 69.

Concatenation

Placing elements in a sequence forms a concatenated chain.

For example, the expression abcd produces the single token abcd.

Escape syntax

You can escape special characters in an expression by preceding them with a backslash or enclosing them in double quotation marks.

For example, both of the following expressions will produce the same token of &&&&&:

- \&{5}
- "&"{5}

The expression [A-Z\-\\\.]{3} will produce any three-character combination of capital English letters, dashes, backslashes, or periods. The following are examples:

- \FC
- -WL
- .KA

Intersections

You use an ampersand to construct an intersection, which produces a token comprised only of values that appear on both sides of the operation.

For example, the expression [A-Za-h]&[a-z] is equivalent to the expression [a-h].

Literals

In the absence of any operators or special characters, Data Governance and Catalog interprets the expression as a literal.

For example, the expression R produces a single token of R.
Note: Data Governance and Catalog interprets regex anchors (^, $, \b, \B, \A , \Z, \z) as literals.

Numeric ranges

You specify numeric ranges by including its endpoints within angle brackets.

For example, the expression <3-6> produces any of the following tokens:

- 3
- 4
- 5
- 6

Note the following limitations when using numeric ranges:

- Include the same number of digits in both endpoints. For example, <000-999> is valid syntax, but <0-999> is invalid syntax.
- Although the expression produces values within a numeric range, it produces them as strings, not integers.
- Numeric range expressions can produce tokens with leading zeros. For example, the expression <000-999> can produce the tokens 000, 009, and 095.
- You can't use negative endpoints.
- The endpoints must fit within the Java int limit. The maximum allowed value is 2147483647.

Optional elements

You can use a question mark to signify that a sub-expression is optional.

For example, the expression Mr(s?) Smith can produce either of the following tokens:

- Mr Smith
- Mrs Smith

Repeated elements

You use a number in curly brackets to indicate how often an expression can output repeated elements.

For example the expression [a-z]{3} can produce the following three-letter tokens:

- iwx
- hxz
- pfm
- ggi

You can produce variable length strings by using digits separated by a comma. This indicates the range of lengths of the tokens.

For example the expression [a-z]{2,4} can produce strings comprised of two, three, or four lowercase English letters. These might include the following tokens:

- py
- hxz
- pfmz
- gg

Unions

Use the pipe character to union together two distinct sub-expressions. This creates an expression that can produce either expression.

For example, the expression id|identification produces either of the following tokens:

- id
- identification

You can use parentheses to shorten the expression. For example, the expression id(""|entification) produces either of the following tokens:

- id
- identification

You can combine any number of elements into a union. For example, the expression oranges|berries|bananas|(red|green) apples produces any of the following tokens:

- oranges
- berries
- bananas
- red apples
- green apples

CHAPTER 8

Data types

Use data access policies with the following data types when you use the Managed Access setting in a delivery template in Data Marketplace or with access policy transformations in Data Integration.

Azure SQL data types

The following table lists the data types that you can use with an Azure SQL data source when using access policy transformations in Data Integration:

Azure SQL data type	Data Access Management data type
bigint	integer
bit	integer
char	string
datetime	timestamp
datetime2	timestamp
datetimeoffset	timestamp
decimal	decimal
float	decimal
geography	unknown
geometry	unknown
hierarchyid	unknown
int	integer
money	decimal
nchar	string
ntext	string
numeric	decimal
nvarchar	string

Azure SQL data type	Data Access Management data type
smalldatetime	timestamp
smallint	integer
smallmoney	decimal
sql_variant	unknown
text	string
time	timestamp
tinyint	integer
uniqueidentifier	unknown
varchar	string
xml	unknown

Azure Synapse data types

The following table lists the data types that you can use with an Azure Synapse data source when using access policy transformations in Data Integration:

Azure Synapse data type	Data Access Management data type
bigint	integer
bit	integer
char	string
date	timestamp
datetime	timestamp
datetime2	timestamp
datetimeoffset	string
decimal	decimal
float	decimal
int	integer
money	decimal
nchar	string
numeric	decimal

Azure Synapse data type	Data Access Management data type
nvarchar	string
real	decimal
smallint	integer
smallmoney	decimal
smalldatetime	timestamp
tinyint	integer
uniqueidentifier	string
varchar	string

Databricks Delta data types

The following table lists the data types that you can use with a Databricks Delta Lake data source when using access policy transformations in Data Integration:

Databricks Delta data type	Data Access Management data type
array	unknown
bigint	integer
char	string
date	date
decimal	decimal
double	decimal
float	decimal
int	integer
map	unknown
smallint	integer
string	string
struct	unknown
timestamp	timestamp

Databricks Delta data type	Data Access Management data type
timestamp_ntz	timestamp
tinyint	integer

Microsoft SQL Server data types

The following table lists the data types that you can use with a Microsoft SQL Server data source when using the Managed Access setting in a delivery template in Data Marketplace and with access policy transformations in Data Integration:

Microsoft SQL Server data type	Data Access Management data type
bigint	integer
char	string
datetime	timestamp
datetime2	timestamp
decimal	decimal
float	decimal
int	integer
money	decimal
nchar	string
ntext	string
numeric	decimal
nvarchar	string
smallmoney	decimal
text	string
varchar	string

PostgreSQL data types

The following table lists the data types that you can use with a PostgreSQL Server data source when using the Managed Access setting in a delivery template in Data Marketplace:

PostgreSQL data type	Data Access Management data type
int2 (smallint)	integer
int4 (integer)	integer
int8 (bigint)	integer
numeric (decimal)	decimal
float4 (real)	decimal
float8 (double)	decimal
varchar	string
text	string
date	date
timestamp	timestamp
timestampz	timestamp
smallserial	integer
money	decimal

Oracle data types

The following table lists the data types that you can use with an Oracle data source when using the Managed Access setting in a delivery template in Data Marketplace and with access policy transformations in Data Integration:

Oracle data type	Data Access Management data type
CHAR	string
DATE	timestamp
DECIMAL	decimal
DOUBLE PRECISION	decimal
FLOAT	decimal
INTEGER	decimal
LONG	string

Oracle data type	Data Access Management data type
NCHAR	string
NUMBER	decimal
NVARCHAR2	string
REAL	decimal
SMALLINT	decimal
TIMESTAMP	timestamp
TIMESTAMP WITH LOCAL TIME ZONE	timestamp
VARCHAR	string
VARCHAR2	string

Snowflake data types

The following table lists the data types that you can use with Snowflake data types:

Snowflake data type	Data Access Management data type
DATE	date
DECIMAL	decimal
INTEGER	integer
TIMESTAMP	timestamp
VARCHAR	string

Available data protections

The following table lists the data protections that you can use with Data Access Management data types:

Data Access Management data type	Description	Available data protections
date	A calendar date	generalize date, redact to null, retain
decimal	A numeric value	redact to null, retain
integer	A positive natural number or zero	regular expression number generator, redact to null, retain
string	A sequence of characters	constant text value, truncate text, regular expression, redact to null, retain

Data Access Management data type	Description	Available data protections
timestamp	The date and time that an event occurred	generalize date, redact to null, retain
unknown	No corresponding data type in Data Access Management	redact to null, retain